A Novel ICESat-2 Bathymetry Model Guided by Photon Density Profiles and Synergy With Sentinel-2 for Automated Seafloor Mapping Without In Situ Data

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***Abstract*—Near-shore shallow-water bathymetry is critical for coastal management, with important scientific and engineering value. Compared to traditional bathymetric approaches, satellite-derived bathymetry (SDB) is more efficient and cost-effective. Especially, the Ice, Cloud and land Elevation Satellite 2 (ICESat-2) carries a photon-counting 532-nm laser capable of penetrating optically shallow waters for bathymetry without requiring in situ measurements. In practical application, heterogeneous benthic substrates and variable water optical properties pose significant out-of-distribution (OOD) challenges to SDB. However, current filtering or clustering models for ICESat-2 SDB, relying on constant configuration and narrow applicability ranges, are unadapted to complex geomorphological conditions. Here, we propose the photon density profile guided cross-modality fusion point cloud segmentation (DECOMPOSE) model for the ICESat-2 SDB mission. Unlike spatially limited conventional approaches, DECOMPOSE achieves comprehensive seafloor characterization by fusing global density profile distributions with microstructural point cloud details. It is resilient against OOD challenges and generalizable to a variety of environments. On a geographically isolated testing dataset, DECOMPOSE outperformed baseline models, with a mean intersection-over-union of 0.9559%, 15.2% higher than the suboptimal comparative model. On a real-world OOD dataset, the DECOMPOSE bathymetric accuracy was excellent, with a root mean square error (RMSE) of 0.4938 m, a mean absolute error (MAE) of 0.2760 m, and a goodness-of-fit R^2 exceeding 0.98. To further improve the usefulness of the proposed paradigm, we integrated DECOMPOSE with a multispectral imagery-based SDB model to establish the data-driven active–passive fusion SDB (DAPF-SDB) model, designed to provide full-coverage underwater topography mapping without in situ data. Quantitative and qualitative analyses demonstrated that, relative to the baseline models, DAPF-SDB achieves accurate and robust seafloor topography mapping under OOD real-world conditions, reducing RMSE and MAE by more than 16% and 14%, respectively. Code for DECOMPOSE and DAPF-SDB is available at https://github.com/qxm1995716/decompose**

***Index Terms*—Cross-modality fusion, deep learning, ICESat-2, photon recognition, satellite-derived bathymetry (SDB), Sentinel-2 multispectral imagery.**

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I. Introduction

SOCIOECONOMIC activity in coastal regions represents more than 60% of the global economic production, with a markedly higher intensity than in continental interior zones [1]. Since near-shore shallow waters constitute essential land–sea transition hubs, their complete bathymetric mapping with high spatial resolution and accuracy is indispensable for sustainable management and coastal regions development. Bathymetric mapping has evolved over millennia, from premeasured heavy ropes mounted on ships, through acoustic echo-sounding to advanced airborne LiDAR bathymetry (ALB). However, existing methods cannot provide cost-effective bathymetry with global coverage and are strongly constrained by local environmental conditions. Conversely, satellite-borne instruments, including passive optical sensors, radars, and, increasingly, LiDARs, provide continuous, near-global remote sensing Earth observations.

In this context, the emerging satellite-derived bathymetry (SDB) is a transformative methodology that leverages the

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enhanced capabilities and spatiotemporal coverage of satellite observations for large-scale, cost-effective underwater terrain mapping. Its fundamental objective is to derive water depths through spaceborne remote sensing using radars, passive optical sensors, or a synergy of those. For indirect bathymetric inversions, conventional SDB methods mainly rely on the spectral reflectance response of the seafloor [2], [3], [4], [5] and the hydrodynamic modulation of sea surface microscale waves by underwater terrain [6], [7], [8]. Since its launch in 2018, the Ice, Cloud and land Elevation Satellite 2 (ICESat-2) of the U.S. National Aeronautics and Space Administration (NASA) has substantially enhanced the applicability of SDB by providing critical observations for direct spaceborne bathymetry. In particular, ICESat-2 carries the Advanced Topographic Laser Altimeter System (ATLAS), the first spaceborne photon-counting lidar for Earth observation. Its central wavelength (532 nm) penetrates optically transparent water bodies, thereby providing direct spaceborne bathymetric measurements and marking a major advance in SDB. Indeed, prior to ICESat-2, observations from the airborne ATLAS prototype, namely Multiple Altimeter Beam Experimental Lidar (MABEL), had undergone extensive bathymetric validation [9], [10]. After the start of ICESat-2 operations, Parrish et al. [11] validated an SDB framework using ATLAS/ICESat-2 Level-2 (L2) Global Geolocated Photon Event Data (ATL03 data product) against in situ ALB measurements, achieving a bathymetric root mean square error (RMSE) of 0.43–0.60 m and a detection capability down to water depths of approximately 40 m.

The fundamental principle of ICEsat-2 bathymetry involves distinguishing photon signals reflected from the surface and seafloor within the observed data, and then calculating water depth based on their elevation difference. Due to this principle, the main difficulty of ICESat-2 SDB is the accurate recognition of target photons from the ATL03 point cloud data. Recent studies mainly assumed preset density characteristics to distinguish photons reflected by the seafloor and at the surface, while alternative approaches have included manual denoising [12]. However, manual extraction is affected by operator subjectivity, resulting in lower stability and consistency than automated approaches. Ma et al. [13] first applied the density-based spatial clustering of applications with noise (DBSCAN) algorithm to the ICESat-2 SDB mission, clustering high-density photon returns from the seafloor and the surface. Xu et al. [14], combining ICESat-2 photons with the Global Surface Water Dataset for lacustrine bathymetry studies, used DBSCAN to identify bottom photons from ATL03 data. To mitigate the sensitivity of DBSCAN to parameter selection, Xie et al. [15] proposed an adaptive optimization methodology that improved recognition accuracy and robustness. The Ordering Points To Identify the Clustering Structure (OPTICS), less sensitive to parameter selection than DBSCAN, was also applied to photon recognition from ATL03 data [16], [17]. As for filtering-based methods, Chen et al. [18] proposed an adaptive variable ellipse filtering bathymetric method, termed AVEBM, that explicitly accounted for photon density attenuation with water depth to enhance noise resilience. Similarly, other studies further improved the accuracy of ICESat-2 SDB, also with elliptical neighborhood filtering, by incorporating horizontal density variations and attenuation priors [19], [20], [21], [22]. Relying on the difference between the density of signals and noises in the ATL03 data, Zhang et al. [23] developed a quadtree clustering algorithm for photon denoising, subsequently modified for seafloor photon extraction [24]. In addition, because surface and seafloor reflections produce distinct photon density peaks in the vertical direction, Jia et al. [25] applied bimodal Gaussian fitting to discriminate between the two signals. He et al. [26] proposed a sliding window-based coastal bathymetric method (SWCBM-Ph), improving the robustness of ICESat-2 SDB against noise and data variability. Currently, machine learning approaches with enhanced generalization and fitting capabilities are increasingly applied to SDB using ICESat-2 data. For example, Meng et al. [27] used neighborhood density and mean distance as input for a backpropagation neural network and derived a robust photon classification. Lin and Knudby [28] proposed an automated approach, using the deep learning-based PointNet++ [29], for ATL03 point cloud segmentation, improving discrimination performance between sea surface and seafloor photons.

Although ICESat-2/ATLAS orbital observation data enable direct spaceborne bathymetry, their sparse spatial distribution, with an approximate spacing of 3.3 km between consecutive ground tracks, precludes full-coverage mapping. Therefore, researchers usually integrate ICESat-2 bathymetry data within semi-empirical/empirical SDB models driven by multispectral imagery to obtain full coverage. Semi-empirical SDB models solve simplified radiative transfer equations to derive mapping functions by calculating a regression curve between satellite remote sensing reflectance and in situ measurements. Empirical SDB models apply statistical learning techniques to establish regression curves between reflectance data and bathymetry without explicit physical prior, offering improved accuracy and generalizability. Le Quilleuc et al. [30] implemented a modified DBSCAN to extract seafloor photons from ICESat-2 data and then used the water depth to calibrate a semi-empirical model and generate accurate bathymetric maps. Cao et al. [31] used the Differential, Regressive, and Gaussian Adaptive Nearest Neighbor (DRAGANN) noise filtering algorithm to obtain ICESat-2 bathymetry measurements, subsequently input into a multiband linear model to derive exclusively spaceborne-based bathymetric maps. Xie et al. [32] conducted a systematic performance evaluation of machine learning-based empirical SDB models trained on bathymetric control points derived from ICESat-2, including support vector regression, random forest, and neural networks. Xi et al. [33] combined ICESat-2 SDB with a band-optimized bidirectional long short-term memory deep learning model to invert water depth from multispectral and hyperspectral remote sensing imagery. More recently, several studies demonstrated the usefulness of ICESat-2 data, in synergy with semi-empirical or empirical SDB models, to enhance the accuracy and robustness of satellite remote sensing-based bathymetric mapping [34], [35], [36], [37], [38].

Although SDB's performance has considerably improved, its accuracy remains insufficient. In particular, cross-regional

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SDB generalization from geographically diverse observations in hydrodynamically complex environments, particularly in near-shore optical shallow waters, remains problematic. Remote sensing optical observations (such as from spaceborne LiDARs and cameras) are sensitive to factors like site-dependent water inherent optical properties (IOPs) and sediment composition. Consequently, geographic cross-regional deployment induces variations in these attributes, thereby generating covariate shifts in observations, which introduces out-of-distribution (OOD) challenges for satellite bathymetric approaches [39]. Besides, because acquiring in situ bathymetry in complex shallow-water environments is difficult, SDB must simultaneously account for the scarcity of bathymetric measurements and allow for robust OOD generalization. However, current clustering and filtering-based ICESat-2 SDB methods, relying on limited data and specific parameter settings, cannot provide the necessary cross-regional generalization capability. In addition, some existing approaches rely on manual selection of segments containing visible seafloor-reflected photon signals during deployment, substantially increasing the time and workload requirements. Moreover, along-track inhomogeneity in seafloor-reflected photon dispersion critically hinders the efficient application of fixed-threshold statistical methods to long ICESat-2 transects, frequently leading to systemic collapse. In particular, signal interferences, such as occlusion-induced gaps or confounded photon signals in very shallow waters, complicate the accurate and continuous identification of surface and seafloor layers. Unfortunately, the cost of expanding the dataset to enhance feature richness and improve the extrapolation capability of SDB models is prohibitive. A more efficient solution is to improve the extraction capability of robust, distribution-invariant global features, at the cost of substantial receptive-field enlargement and a considerable increase in computational complexity, which precludes practical implementation.

To mitigate these constraints, this study proposes a novel ICESat-2 SDB model that enhances bathymetric accuracy and cross-regional transferability by implementing a complete, fully automated bathymetry mapping algorithm. In particular, a novel photon density profile guided cross-modality fusion point cloud segmentation (DECOMPOSE) model is proposed using deep learning. Compared with existing single-modality algorithms, DECOMPOSE decouples input ICESat-2 ATL03 photons into dual modalities: density imagery and point cloud, to capture multiscale photon features. It introduces a cross-modality fusion paradigm that integrates representations from two complementary views of the data. Density profiles capture robust large-scale photon distribution patterns, whereas photon point cloud clusters preserve fine-scale structural information. Together, these synergistic representations expand the receptive field beyond conventional methods without sacrificing detail. To compensate for ICESat-2's sparse geographic coverage, we further combine DECOMPOSE with Sentinel-2 multispectral imagery using the multiple scale resolution fusion model (MuSRFM) [40]. This integration leads to an innovative, fully spaceborne bathymetric retrieval paradigm that does not require in situ data and enables full-coverage underwater topography mapping, referred to as the data-driven active–passive fusion SDB (DAPF-SDB) model. To account for complex practical environments and regional variations in attributes such as water IOPs and sediments, a geographically isolated testing dataset is employed to validate the photon recognition performance of DECOMPOSE. In addition, a more extensive real-world dataset is employed to evaluate the cross-regional generalizability of both DECOMPOSE and DAPF-SDB. A thorough analysis establishes DECOMPOSE's distribution-invariant, cross-scale bathymetric robustness across diverse environments, and also confirms that DAPF-SDB enables accurate, full-coverage underwater topography mapping without in situ data.

II. Datasets and Study Area

A. Data

1) ICESat-2 ATL03 Point Cloud Data: ICESat-2 is the first Earth-observing satellite mission to carry a photon-counting spaceborne LiDAR, providing an unprecedented view for studies of atmosphere, glaciers, and oceans [41]. The primary data source for this study is the ICESat-2 L2 ATL03 geolocated point cloud data product, which contains ATLAS captured photon events after geophysical calibration. In preprocessing, a decoding script is deployed to extract geographic coordinates, elevation, segment indices, and ancillary metadata. Using the decoded data, along-track distances are calculated, and the original data are converted to the Universal Transverse Mercator (UTM) system to obtain a processed photon point cloud. Because ICESat-2 SDB is fundamentally a photon point cloud semantic segmentation mission, further processing is required to convert the recognition results into bathymetry. Thus, the ground truth (GT) format is ultimately dictated by its application scenario. In particular, for supervised learning and photon recognition performance assessment, GT for the training/testing dataset comprises manually annotated semantic labels classifying photon events into noise, seafloor, and surface (both sea and land surface) classes. For bathymetric assessment, GT for the real-world dataset is interpolated from a digital elevation model (DEM) derived from ALB data of the digital coast project (https://coast.noaa.gov/digitalcoast) that managed by the U.S. National Oceanic and Atmospheric Administration (NOAA). These elevations are converted into bathymetry using the DTU21 mean sea surface (MSS) [42], so that their vertical datum aligns with the tide correction of ICESat-2 SDB based on the DTU23 model [43].

2) Sentinel-2 Multispectral Imagery: Due to the spatial constraint imposed by the sparsity of ICESat-2 ground tracks on its bathymetric capability, we introduce the DAPF-SDB active–passive fusion model to integrate the ICESat-2 ATL03 point cloud data with multispectral remote sensing imagery. For this purpose, atmospherically corrected Sentinel-2 L2A surface reflectance (SR) imagery, released by the European Space Agency (ESA), serves as a secondary data source. To eliminate transient noise from vessel wakes and sun glint that degrades SR fidelity, stable SR composites are generated via temporal median filtering [40], [44]. This step is implemented on Google Earth Engine to enhance efficiency, leveraging its cloud-based parallel processing capabilities for large-scale

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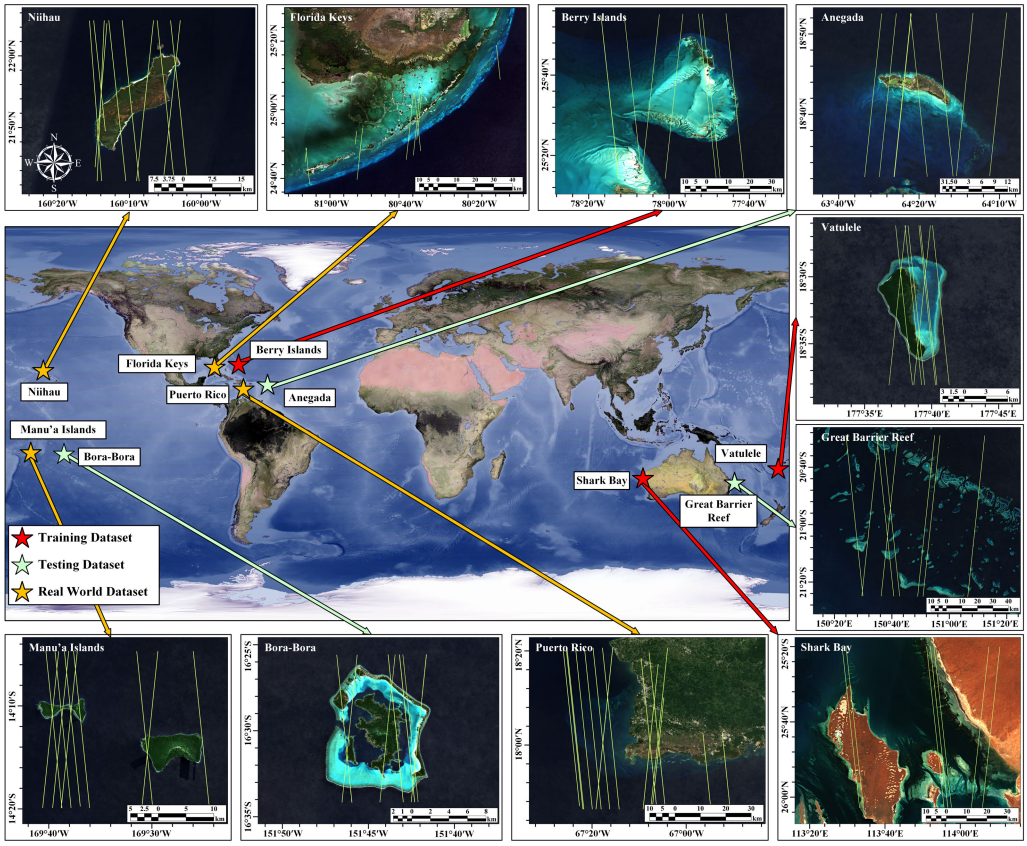


Fig. 1. Geographic distribution of near-shore shallow-water sites used in this study.

spatiotemporal data. Bilinear interpolation is then employed as a preprocess on null-value pixels to prevent contamination of gradient-based backpropagation by anomalous values and ensure stable model training.

B. Study Area

In this study, we select ten near-shore optical shallow-water sites to constitute the training, testing, and real-world datasets (geographic distribution in Fig. 1). The Berry Islands, Shark Bay, and Vatulele constitute the training dataset; Anegada, Bora–Bora, and part of the Great Barrier Reef constitute the testing dataset; and the remaining sites constitute the real-world dataset. These geographically isolated sites provide essential data for this study, with spatial independence and distinct functional roles. In particular, the training dataset provides supervision for model training, the testing dataset assesses photon recognition performance, and the real-world dataset validates bathymetric performance for OOD environments. Although the experimental sites are primarily in tropical and temperate regions, representing a nonglobal dataset, their distribution across different oceans and geomorphic settings (e.g., islands and coastlines) still allows a rigorous assessment of bathymetric generalization with reduced geographical bias.

1) Training and Testing Datasets: Each dataset comprises ATL03 data from 24 ICESat-2 tracks in three spatially isolated regions, respectively. In particular, the training and testing

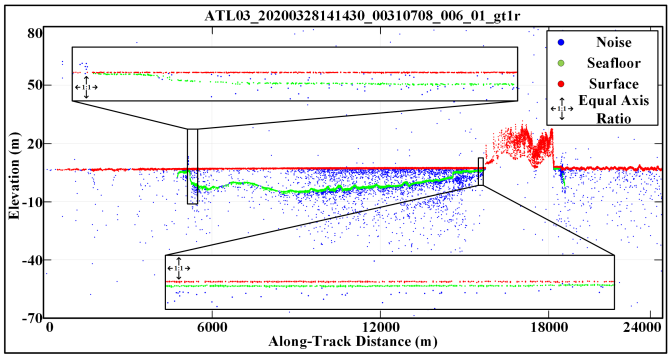


Fig. 2. Illustration of a manually labeled point cloud from the ATL03_20200328141430_00310708_006_01_gt1r scan line, a 24-km ground track transect from the Ice, Cloud and Land Elevation Satellite-2 (ICESat-2) over the Vatulele lagoon.

datasets are used to calibrate and evaluate the photon semantic segmentation model; thus, we manually annotated both datasets with discrete semantic labels and incorporated these labels into the GT specifications, classifying each photon as noise, seafloor, or surface. According to quantitative statistics, the training and testing datasets contain approximately 650 000 and 500 000 photons, respectively, with the proportions of noise, seafloor, and surface photons being 25:16:59 in the training dataset and 38:7:55 in the testing dataset. As observed, the proportion of surface reflected photons is comparable between the two datasets, whereas the propor-

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TABLE I
DETAILS OF THE SENTINEL-2 REMOTE SENSING DATA

Site	Data Acquisition Period
Niihau	20190101——20230101
Manu'a Islands	20200101——20240101
Florida Keys	20160101——20220101
Puerto Rico	20170101——20210101

tion of seafloor photons decreases markedly in the testing dataset, indicating a substantial difference in their compositional characteristics. Fig. 2 shows a manually annotated ICESat-2 transect capturing primarily lagoon topography near Vatulele. It can be observed that the high-density clusters below the surface indicate seafloor-reflected photon events. The observed seafloor returns in Fig. 2 indicate a strong anticorrelation between photon event density and water depth, while the density of noise photons is substantially higher near the coast. To increase fidelity and minimize the number of false positives, photon events closest to the seafloor but with anomalously low density were annotated as a noisy signal. In fact, the morphology of the ICESat-2 seafloor-return photon point cloud depends on multiple factors, including underwater topography and atmospheric noise. Thus, the photon point cloud frequently appears fragmented, i.e., composed of discontinuous layers, because of obstructions or water-depth variations. Furthermore, biological groups, suspended matter, water IOPs, and benthic properties can generate massive noise events in the observations, complicating photon recognition.

2) Real-World Dataset: As stated previously, the response to the ICESat-2 incident laser signal is influenced by the water column and seafloor optical properties, themselves directly related to the geographical location. Consequently, intersite variability can introduce data distribution shifts in satellite observations, leading to a prototypical OOD scenario featured by divergent topography, dynamic sea surface morphology, and distinct noise patterns. To capture this OOD challenging scenario, four geographic sites are selected as a real-world dataset (locations in Fig. 1). We extracted 10, 10, 11, and 16 ICESat-2 transects for Niihau, the Manu'a Islands, the Florida Keys, and Puerto Rico, respectively. Compared with the training and testing datasets, the real-world dataset exhibits greater complexity and a broader spatial distribution, inducing notable pattern variations. Therefore, the real-world dataset is suitable to assess the model's cross-regional generalizability. Furthermore, the real-world dataset is also used to evaluate the full-coverage shallow-water seafloor mapping without relying on in situ data. Accordingly, temporally median-fused Sentinel-2 L2A SR imagery is incorporated in addition to ICESat-2 photon point cloud, and Table I summarizes the Sentinel-2 imagery used for the four regions in the real-world dataset. Notably, ICESat-2 transects were chosen according to data quality, since sparse coverage and long revisit period restrict data availability, yielding an uneven spatial distribution that challenges MuSRFM inference in DAPF-SDB.

In contrast to the testing dataset, which employs manually annotated semantic labels as GT to evaluate photon recognition performance, the real-world dataset uses in situ DEM to assess the bathymetric performance of DECOMPOSE and DAPF-SDB. Following the data description, the DEM was converted into a Digital Bathymetry Model (DBM) using DTU21 MSS as the datum. DECOMPOSE bathymetry is evaluated using point-level GT, obtained by interpolating the DBM at individual seafloor photon locations, whereas DAPF-SDB bathymetry uses area-level GT represented by the full DBM grid over the corresponding region.

III. METHOD

Existing ICESat-2 SDB approaches constrain the receptive field to geometrically fixed neighborhoods in methods relying on circular or elliptical search areas, such as DBSCAN and AVEBM, or PointNet++, that use a predetermined small-scale input. However, complex ICESat-2 point cloud morphologies generated by occlusion, noise, underwater topography, or water IOP variability are challenging for photon recognition models dependent on localized sampling regimes. The proposed model, DECOMPOSE, efficiently enlarges the receptive field while retaining fine-scale structural information. As shown in Fig. 3, DECOMPOSE integrates three main components. An imagery modality branch is designed to process density profiles, while a point cloud modality branch handles the photon subsets. These two modalities are then fused by an integration module through an innovative cross-modality information synthesis (C-MIS) mechanism. This fusion enables the joint extraction of large-scale distributions from density profiles and fine-scale local structures from point clouds. Building on this, DECOMPOSE was integrated with a Sentinel-2 L2A SR imagery-based SDB model, MuSRFM, to form DAPF-SDB. By integrating active and passive SDB methods, this fusion extends DECOMPOSE's point-scale sparse measurements to area-scale, full-coverage bathymetric mapping, without reliance on in situ data.

A. Imagery Modality Branch

1) Photon Density Profile Image Generation: The imagery modality branch produces photon density profile images from ICESat-2 observations, leveraging their robust representation of large-scale photon distributions. The photon density profile here is defined as a 2-D density map of photon signals in the distance–elevation plane, as illustrated in Figs. 3 and 4(a). The photons projected onto the distance–elevation plane are sorted into a regular grid [see Fig. 4(a)] using a 2-D sliding window with Δd = 2.5 m and Δe = 0.25 m, and a stride equal to the window size in both directions (nonoverlapping windows). These settings, determined empirically through comparative analyses, balance the capability to resolve photon density variations with the horizontal–vertical photon distribution characteristics. The photon count (pixel value) in each grid cell defines the photon density. Once the sliding window has covered the entire ICESat-2 track, a photon density profile image is generated. To minimize contamination by unfixed photon count ranges and ensure stability, pixel values are normalized as follows:

$$ndp(x,y) = \frac{dp(x,y)}{max(dp)} \tag{1}$$

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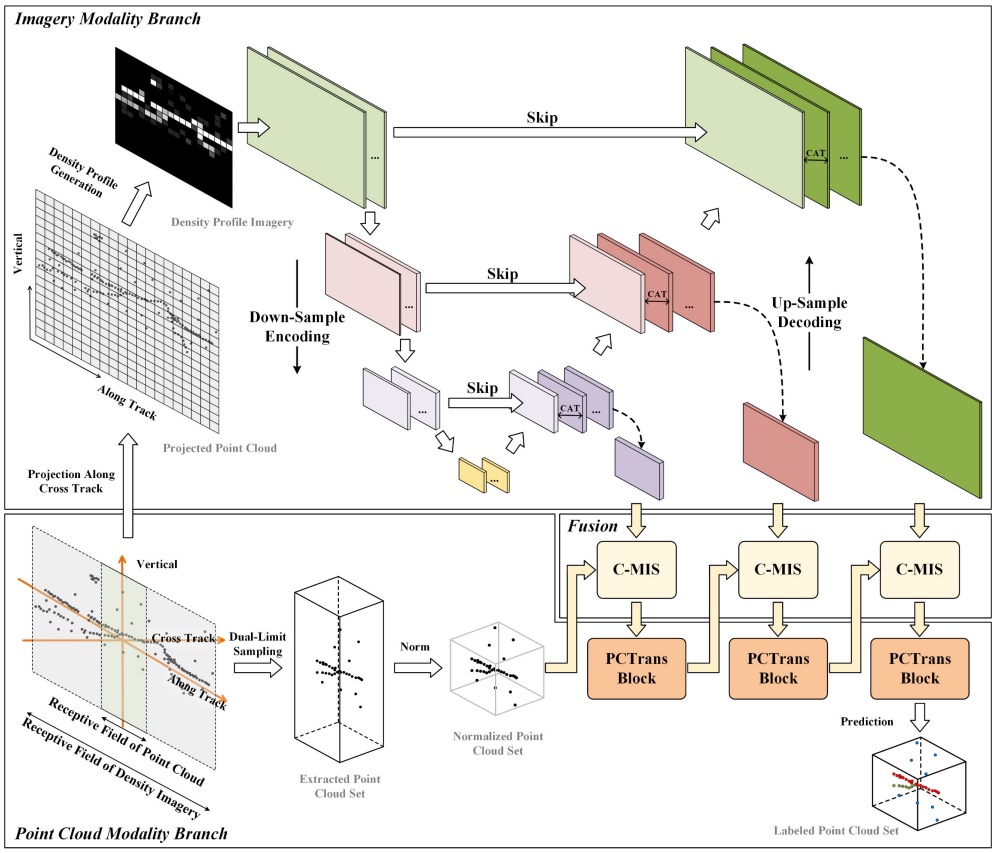


Fig. 3. Overall structure of the proposed DECOMPOSE model showing the imagery modality branch, the point cloud modality branch, and the integration module relying on C-MIS.

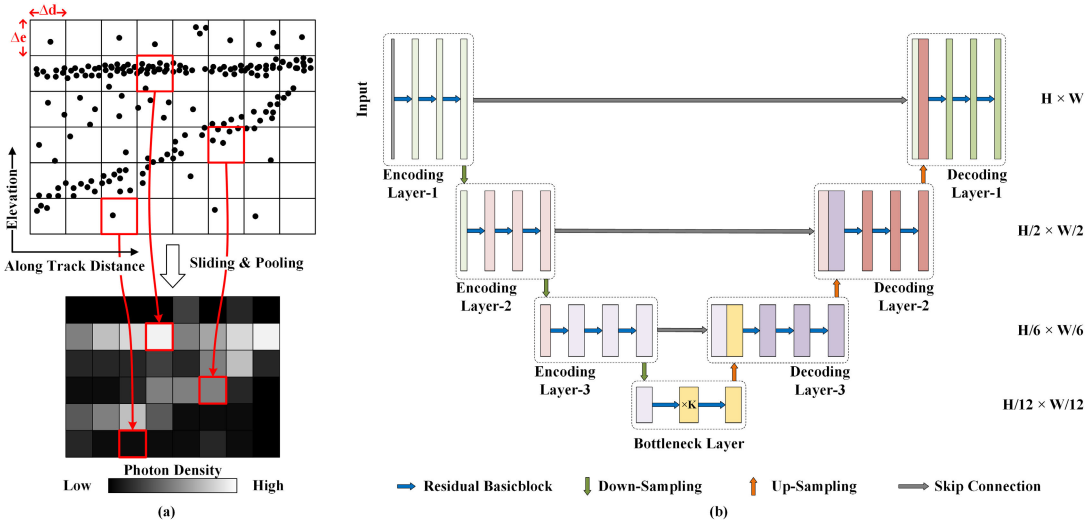


Fig. 4. (a) Generation of photon density profile images from ICESat-2 measurements. (b) Schematic representation of the U-Net model used as a backbone for the imagery modality branch.

where *ndp* is the normalized density profile, *dp* is the original density profile, and (x, y) are pixel coordinates. Normalization restricts the photon density values to [0,1]. As illustrated in Fig. 4(a), the calculated density profiles efficiently characterize global photon distributions with a few pixels and can be used for intuitive diagnostics. However, despite the receptive-field expansion and improved model capability to capture broader contextual patterns, pooling introduces spatial blurring that results in the loss of fine-scale structural details.

2) U-Net Model With Photon Density Profiles as Input: To capture robust macroscopic features, a U-Net model [45] [see Fig. 4(b)] is introduced as the backbone of the imagery modality branch with photon density profile images as input. U-Net is a well-established encoder–decoder network that

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TABLE II
PSEUDO-CODE FOR THE DUAL-LIMIT PHOTON POINT CLOUD SAMPLING ALGORITHM

Input: Raw ICESat-2 photons Ps, preset along-track distance threshold PD and preset photon count threshold PN, valid photon ratio Rv.
Initialization: Photon index $i = 0$, sampled sets $\boldsymbol{S} = \emptyset$, $ATD(\cdot)$ is the calculation of along-track distance.
Sort the Ps by along-track distance.
While $i < Len(Ps)$:
  $k = \min[i + PN - 1, Len(Ps)]$;
  If $ATD(p_k) - ATD(p_i) \le PD$:
    $\boldsymbol{S} \cup \{\{p_i, p_{i+1}, p_{i+2}, p_{i+3}, \dots, p_k\}\}$;
    $i = k + 1$;
  Else:
    $t = i$;
    While $t < Len(Ps)$ & $ATD(p_t) - ATD(p_i) \le PD$:
      $t \mathrel{+}= 1$;
    If $\frac{(t-i+1)}{PN} > Rv$:
      $\boldsymbol{S} \cup \{\{p_i, p_{i+1}, \dots, p_t, \underbrace{np_{t+1}, np_{t+1}, \dots, np_{PN}}_{padded\ placeholders}\}$;
    $i = t + 1$;
return: $\boldsymbol{S}$;

progressively expands its receptive field through stacked convolutions and downsampling, while recovering fine-scale spatial details using skip connections and upsampling layers, effectively achieving multiscale feature fusion. In DECOMPOSE, a four-layer U-Net employing average pooling for downsampling and bilinear interpolation for upsampling is implemented [the feature map shape in each layer is illustrated in Fig. 4(b)]. The encoding and decoding chains are both constituted by residual basic blocks [46] to ensure stability and robustness. Notably, there is no GT for this U-Net model. Instead, its output multiscale feature maps are fed into the fusion module based on C-MIS to derive photon distributions at multiple scales that cannot be obtained from localized photon point cloud subsets.

B. Point Cloud Modality Branch

1) Dual-Limit Photon Point Cloud Sampling: Because the length of a single ICESat-2 transect can reach several tens of kilometers, directly processing the associated massive ATL03 point cloud is computationally prohibitive. Therefore, the input point cloud data must be partitioned into segments to reduce computational complexity. However, sampling point cloud segments using fixed spatial intervals results in variable photon counts because of the heterogeneous along-track photon density, yielding inconsistent photon counts between sampled clusters. Conversely, the heterogeneous along-track photon density can also result in inconsistent spatial intervals between clusters if the sampling strategy assumes a fixed photon count. The former produces sampled clusters with variable volumes, whereas the latter method results in unmatched spatial ranges. Consequently, in both cases, the sampled photon point cloud clusters cannot be correctly normalized, thereby degrading the downstream photon recognition performance. Therefore, to avoid spatial distortion, a dual-limit sampling strategy is adopted, simultaneously constraining both spatial range and photon count of the sampled point cloud clusters, as described in Table II. During this process, variation of density may result in sampled clusters containing very few valid photons, causing them to be dominated by padded placeholders and consequently lose their significance. Hence, a parameter named valid photon ratio (R_V) is introduced in this study and fixed at 0.1, to eliminate clusters with an excessively low proportion of photons while balancing data-content richness and overall data volume. Following Lin and Knudby [28], the extracted point cloud samples are represented as four-channel tensors with dimensions easting ($\mathrm{UTM_X}$), northing ($\mathrm{UTM_Y}$), elevation, and decoded oceanic confidence. These tensors are then used as input for the point cloud modality branch. For computational stability, the coordinates of sampled photons in each cluster are normalized to a fixed range (typically [0,1]) as follows:

$$np_i = \frac{p_i}{s_i}, \quad p_i \in sampled\ set \tag{2}$$

where p and np are the photon spatial coordinates before and after the normalization process, respectively, s indicates the chosen normalization range, and i represents the channel index. This normalization is applied exclusively to the spatial coordinates (after conversion to the UTM coordinate system), including $\mathrm{UTM_X}$, $\mathrm{UTM_Y}$, and elevation. Fragments with insufficient photon counts are padded with a constant value of -111.0 to preserve consistency of the sampled volumes.

2) Transformer With Point Cloud Samples Used as Input: The Transformer deep learning model [47], with its self-attention originally designed for sequential data, is intrinsically suited to process permutation-invariant structures such as point clouds [48], [49], [50]. Therefore, a point cloud Transformer (PCTrans) model is adopted as the backbone of the point cloud modality branch. This branch includes three PCTrans blocks, each comprising multiple serially connected encoding units. Inputs to each PCTrans block arise from the associated C-MIS unit (see Fig. 3) as tensors with size $N \times C$ (N samples with C channels). Within the encoding unit, the input tensor first undergoes layer normalization and then is processed by a multiple-layer perceptron (MLP) to generate Query (Q), Key (K), and Value (V) matrices. A multihead self-attention (MHSA) module further partitions the derived Q, K, and V matrices into feature subspaces (or heads) and computes the scaled dot-product attention for all heads in parallel

$$y_i = softmax\left(\frac{Q_i \times K_i^T}{\sqrt{C}} + M\right) \times V_i. \tag{3}$$

Here, y_i is the output sequence, M is the mask used to identify placeholders for padding, and i is the feature subspace index. As for the mask M, a binary mask M_V of length N matching that of the input sequence is first generated with values of 0 and 1 assigned to placeholders and valid elements, respectively, and then the mask M is derived from M_V as

$$M = M_V \times M_V^T \tag{4}$$

$$M_{j,k} = \begin{cases} -111.0, & if\ M_{j,k} == 0 \\ 1, & if\ M_{j,k} == 1 \end{cases}, \quad \forall j,k \in [0, N). \tag{5}$$

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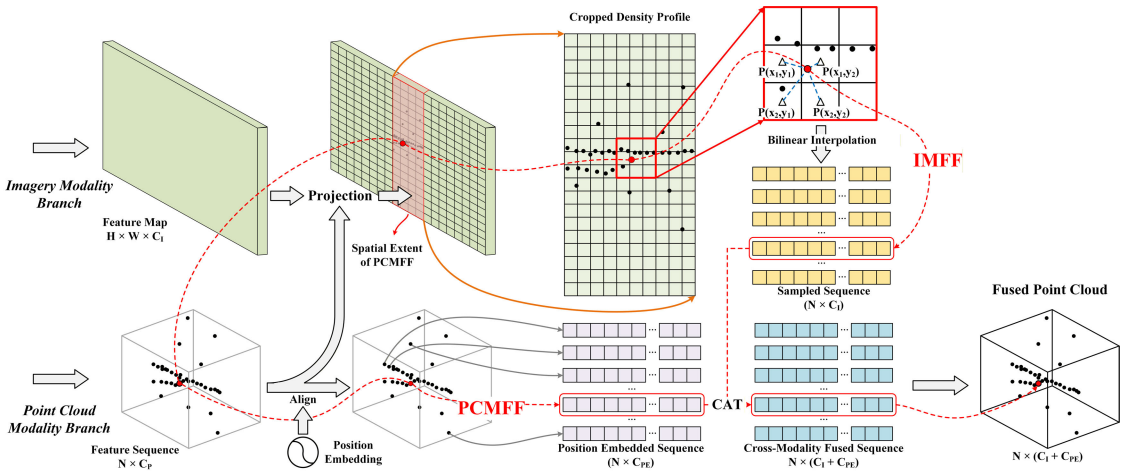


Fig. 5. Workflow of a single C-MIS unit within the DECOMPOSE fusion module.

MHSA output subfeatures are concatenated channel-wise and then processed by projection layers, feedforward networks, and residual operations to produce the output of the encoding unit.

3) Prediction Head: To convert the PCTrans block output into categorical labels, a lightweight prediction head, attached to the point cloud modality branch, projects high-dimensional feature sequences onto a set of one-hot semantic labels. In particular, the prediction head consists of two MLP layers that map the feature representations to class-specific logits, followed by a logarithmic softmax to obtain probability vectors

$$y_i = \log\left(\frac{e^{x_i}}{\sum_{j=1}^{K} e^{x_j}}\right) = x_i - \log\left(\sum_j e^{x_j}\right) \tag{6}$$

where y_i is the predicted probability for class i, x_i is the corresponding logit value generated by the head, and K is the total number of classes. One-hot semantic labels are finally obtained for each channel by applying the argmax function to the corresponding probability vector.

C. C-MIS-Based Integration Module

The core innovation of DECOMPOSE is its efficient cross-modality fusion strategy, realized through three C-MIS units (see Fig. 3) that effectively fuse complementary features from both modalities. This strategy efficiently alleviates modality-specific representation disparities by integrating macroscopic photon distributions derived from density profiles with microscopic structural features derived from point cloud subsets. The computational workflow of a single C-MIS unit is illustrated in Fig. 5. Due to the dual-modality architecture of DECOMPOSE, each C-MIS employs two parallel feedforward paths. In detail, the imagery modality feedforward flow (IMFF) processes feature maps from the imagery branch with dimensions $H \times W \times C_I$, while the point cloud modality feedforward flow (PCMFF) processes feature sequences from the point cloud branch with dimensions $N \times C_P$.

Operating at photon event level, the IMFF first includes an elementwise projection of the point cloud data (with each point defined by its along-track distance and elevation) onto density profile feature maps to characterize the geolocation of each photon within the density profile domain. As illustrated in Figs. 3 and 5, PCMFF operates with a limited receptive field due to sparse photon events. In contrast, IMFF achieves large-scale perception through density profile images derived from ICESat-2 transects, underscoring the essential role of density profiles in comprehensive seafloor analysis. For each projected photon, a feature vector is extracted via bilinear interpolation on the density profile derived feature map, yielding a photon-wise representation that situates each photon within the larger-scale context. Fig. 5 illustrates an example photon event (highlighted in red), for which feature extraction is performed by bilinear interpolation among the four nearest neighbors surrounding the projected photon location, calculated as

$$\begin{aligned} P(x,y) = {} & P(x_1,y_1) * (x - x_1) * (y - y_1) \\ & + P(x_1,y_2) * (x - x_1) * (y_2 - y) \\ & + P(x_2,y_1) * (x_2 - x) * (y - y_1) \\ & + P(x_2,y_2) * (x_2 - x) * (y_2 - y) \end{aligned} \tag{7}$$

where x and y are the along-track distance and elevation of the target photon, and (x_1, x_2) and (y_1, y_2) are the pairs of horizontal and vertical coordinates, respectively, of its nearest neighbors (see Fig. 5, upper right) in the imagery domain. By iteratively applying this interpolation to all photons, feature sequences of dimensionality $N \times C_I$ are obtained, representing large-scale distributions and achieving modality alignment.

In contrast, the PCMFF algorithm is more streamlined. The input from the point cloud modality branch is first projected through an MLP to align its channel dimensions with those of the positional embedding sequence. In contrast to the 3-D coordinates of DECOMPOSE's point cloud input, positional embedding in this branch adopts a 2-D spatial representation from the photon density profile, thereby implicitly capturing spatial position features at the profile scale. In particular, positional embedding in this branch is performed by elementwise summation of point-feature sequences and positional vectors

$$y = f_a(x) + PE \tag{8}$$

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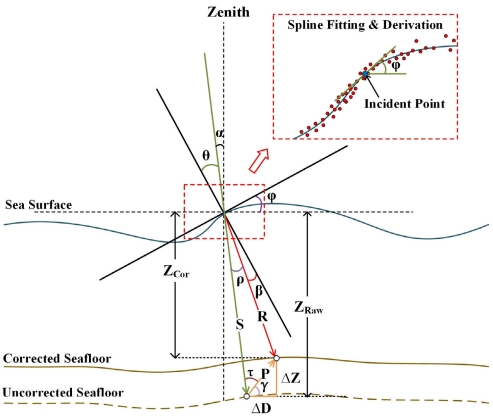


Fig. 6. Schematic of the refraction correction.

where y is the positionally embedded PCMFF output feature, x is the PCMFF input, f_a is the MLP used for alignment, and PE represents position embedding vectors and is obtained by a mapping function f_{pe} with the along-track distance and elevation as input

$$PE = f_{pe}\left([dist_{at}, elev]\right). \tag{9}$$

Finally, the IMFF and PCMFF outputs are concatenated to produce fused features with dimensionality $N\times\ (C_I + C_{PE})$. Combining the IMFF and PCMFF within a C-MIS unit removes modality-specific heterogeneity, allowing DECOMPOSE to extract comprehensive representations. As shown in Fig. 3, the integration module includes three C-MIS units associated with different U-Net levels in the imagery modality branch. The first C-MIS unit introduces large-scale abstract representations of the photon density profile, allowing DECOMPOSE to capture global photon distributions across extensive spatial scales and to establish a foundational semantic context. Following C-MIS units introduce finer photon distribution information from the density profile domain to gradually refine the model output.

D. Refraction and Tidal Corrections

Because DECOMPOSE provides photon recognition results rather than direct water-depth measurements, corrections for refraction and tide are critical postprocessing steps. In principle, refraction correction compensates positional deviations caused by light refraction at the atmosphere–water interface. In this study, the correction method proposed by Chen et al. [51] is adopted, which accounts for both incident angle and sea surface fluctuations (see Fig. 6). First, sea surface photons are extracted within a fixed spatiotemporal range based on the transmission time and geolocation of all seafloor photons. Then, the local sea surface slope φ at the incident point is determined with a spline function, while Snell's law is applied to the zenith angle α to derive the corresponding incident angle θ, as expressed in (10), the refraction angle β, and the true light-ray length R. Similarly, the geometric deviation angle γ of seafloor photon is determined from φ, α, and τ as expressed in (11)

$$\theta = \begin{cases} \alpha - \varphi, & \alpha > \varphi \\ \varphi - \alpha, & \alpha \le \varphi \le \alpha + \pi/2 \\ \pi - \varphi + \alpha, & \alpha + \pi/2 < \varphi \le \pi \end{cases} \tag{10}$$

$$\gamma = \begin{cases} \pi/2 + \alpha + \tau, & \alpha > \varphi \\ \pi/2 + \alpha - \tau, & \alpha \le \varphi \le \alpha + \pi/2 \\ \pi/2 + \alpha + \tau, & \alpha + \pi/2 < \varphi \le \pi. \end{cases} \tag{11}$$

Subsequently, the horizontal (ΔD) and vertical (ΔZ) seafloor photon offsets are derived from the positional discrepancy P (see Fig. 6) between the coordinates before and after refraction correction. The horizontal offset ΔD is then decomposed into northing and easting components using the azimuth angle κ

$$\Delta N = \Delta D * cos\left(\kappa\right) \tag{12}$$

$$\Delta E = \Delta D * sin\left(\kappa\right). \tag{13}$$

Finally, refraction-corrected seafloor photons are obtained by applying calculated offsets.

Following the refraction correction, bathymetric values can be calculated by adding tide-induced elevation variations to the elevation difference between the surface and the corrected seafloor. Here, the tidal correction ΔE_T is calculated with the DTU23 model, with the photon position and transmission time as input. To determine the sea surface elevation, photons reflected by the sea surface corresponding to each recognized seafloor photon are first identified, and the sea surface elevation E_S is then obtained by interpolating between these determined sea surface photons. Finally, the bathymetry Z is calculated as

$$Z = E_S + \Delta E_T - E_W \tag{14}$$

where E_W is the refraction-corrected seafloor photon elevation.

E. DAPF-SDB Model

Although ICESat-2 active remote sensing measurements are clearly useful for SDB, their operational utility for scientific and engineering applications is restricted by the broad spatial interval (3.3 km) between ICESat-2 ground tracks and the long revisit time. Therefore, studies using ICESat-2 data commonly integrate it within passive multispectral imagery-based SDB models, such as the logarithmic band ratio (LBR) model or random forest model, to extrapolate the discrete ground tracks to continuous spatial coverage. However, such models, which rely on limited receptive fields (e.g., pixel-wise mapping), remain susceptible to interference caused by heterogeneity in the environmental parameters. Moreover, the sparse and irregular bathymetric control points derived from ICESat-2 lack representativeness, further complicating regression analyses.

To address the representativeness problem, the proposed DECOMPOSE model is integrated with the Sentinel-2-based MuSRFM into a synergistic framework, termed DAPF-SDB. This framework relies exclusively on spaceborne observations, without in situ measurements, and establishes a novel paradigm for SDB missions. As a multiple-scale resolution fusion-based model, MuSRFM outperforms conventional

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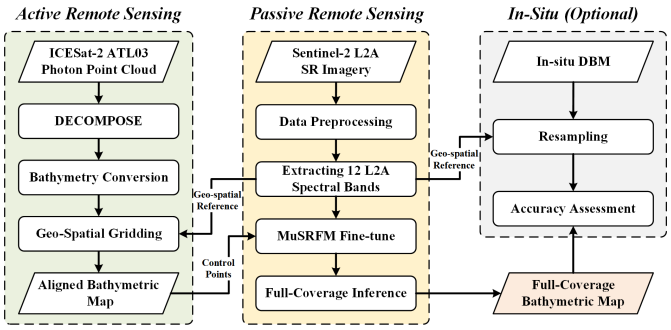


Fig. 7. Schematic of the proposed DAPF-SDB paradigm established to extend the ICESat-2 spatial coverage.

semi-empirical or empirical SDB models both for accuracy and generalizability. It is particularly effective for spatially complex environments where optical parameter homogeneity cannot be assumed. Notably, MuSRFM achieves substantial gains for extrapolation from sparse control points, thus demonstrating its compatibility with the ICESat-2 sampling characteristics. Fig. 7 details the DAPF-SDB workflow, with three main components: an active remote sensing module for ICESat-2 data, a passive remote sensing module for Sentinel-2 data, and an optional validation module to assess bathymetric accuracy.

In particular, DECOMPOSE is first employed for ICESat-2 photon recognition, and bathymetric points are then derived by applying refraction and tidal corrections. After that, geo-spatial gridding is applied to align the point-wise bathymetry with Sentinel-2 SR imagery. A raster grid fully consistent with the geo-spatial reference of the input Sentinel-2 SR imagery is constructed. Then the bathymetric point set is traversed to identify raster cells containing valid measurements. For each raster cell, the K nearest bathymetric points relative to the cell-center coordinates are extracted, and the cell depth is estimated using inverse distance weighting (IDW) interpolation based on these points, that is,

$$A(s) = \frac{\sum_{i=1}^{K} Z(i)/dist(s,i)}{\sum_{i=1}^{K} 1/dist(s,i)}. \tag{15}$$

Here, A is the aligned bathymetric map, s is the cell containing valid measurements, Z is the bathymetric point set, and $dist(\cdot)$ is the Euclidean distance function. For passive remote sensing, the initial step involves preprocessing the input Sentinel-2 L2A imagery, primarily by inpainting invalid pixels across all bands. Subsequently, the pretrained MuSRFM within DAPF-SDB is fine-tuned with DECOMPOSE-derived water depths, thereby adapting its weights to the target data distribution. MuSRFM is designed to explicitly use all 12 Sentinel-2 L2A bands as input for water-depth estimation, and the same input configuration is adopted here to ensure consistency. Fine-tuning supervision is provided by the aligned bathymetric map, which is strictly co-registered with the input Sentinel-2 imagery. Since ICESat-2 transects are spatially sparse, the integrated MuSRFM samples every valid pixel in the bathymetric map as a center to sample patches to maximize the scale of the supervisory dataset. The fine-tuned MuSRFM then extrapolates to the full study area, completing the DAPF-SDB workflow. The valid bathymetric scale is delineated using in situ DBM or, when unavailable, constrained by shoreline distance and water-depth thresholds. In situ measurements are used solely for accuracy evaluation and are never used in any stage of model weight optimization.

IV. Experimental Results

A. Experimental Configuration

1) Model Settings: The U-Net within the DECOMPOSE imagery modality branch comprises four layers with 4, 4, 4, and 5 residual basic blocks and 32, 64, 128, and 128 channels, respectively, from top to bottom. The decoder architecture is symmetrical. The three PCTrans backbone blocks within the point cloud modality branch comprise three encoding units each, and each unit includes a four-head MHSA module. For model optimization, the negative log-likelihood (NLL) loss function is adopted as the optimization goal and is defined as

$$NLL\ Loss(x,y) = -\frac{1}{N}\sum_{i=1}^{N} p(y_i|x_i) \tag{16}$$

where x and y are the prediction and the GT, respectively, N is the number of samples, and $p(y_i|x_i)$ is the logarithmic softmax output for sample i. Model training, conducted over 15 epochs, combined the AdamW optimizer [52] with a cosine annealing scheduler; the learning rate value was initialized at 10^{-3} with a minimum of 10^{-6}. For DAPF-SDB, the inference results from DECOMPOSE are used to fine-tune the pretrained MuSRFM. To fully exploit its prior knowledge, the learning rate is set to $1e^{-4}$ with ten training epochs, while all the remaining settings follow the default configuration.

To provide a comprehensive comparison, two groups of baseline models were considered for the two missions in this study. For the photon recognition mission, three models were selected: DBSCAN [13], AVEBM [18], and PointNet++ [28]. DBSCAN, the only unsupervised clustering algorithm among the three baselines, outputs cluster indices (i.e., integer labels for photon assignment to clusters) instead of a semantic category. Therefore, for each cluster generated by DBSCAN on the annotated testing dataset, the annotated semantic labels of photons belonging to that cluster were tallied, and the majority label was assigned to photons within the cluster. Consequently, DBSCAN cannot be applied to real-world scenarios lacking manual annotations. Regarding hyperparameters, DBSCAN used a minimum sample threshold of 3 and a neighborhood radius of 1.5 m, while the other models were initialized with default settings. For the full-coverage bathymetry mission, the classical LBR semi-empirical model [3] and the commonly used machine learning-based empirical models, including Random Forest Regressor (RFR) [53] and CatBoost [54], were employed as benchmarks. In addition, to assess the contribution of DECOMPOSE-derived control points in DAPF-SDB, MuSRFM was directly extrapolated without fine-tuning, and its accuracy was statistically analyzed for comparison. Here, LBR employs a scale constant of 15 000. RFR adopts 200 estimators, with a minimum sample size of 2 enforced at both internal split nodes and leaf nodes, whereas CatBoost is set to

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a depth of 100. For both machine learning models, the input spectral bands are B1–B5 and B11 [54].

2) Quantitative Evaluation: Quantitative performance was evaluated by computing corresponding metrics under two scenarios with adapted validation source: annotated labels for photon recognition, and in situ water depths for bathymetry.

For photon recognition evaluation, overall accuracy (OA), precision, recall, $F1$-score, intersection over union (IoU), and mean IoU (MIoU) were used. The OA, precision, and recall are defined as

$$OA = \frac{\sum_{i=1}^{K} TP_i}{N} \tag{17}$$

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{18}$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \tag{19}$$

where K is the number of photon classes, N is the number of samples, and TP_i, FP_i, and FN_i are the number of true positives, false positives, and false negatives for class i. The F1-Score is defined as the harmonic mean of precision and recall

$$F1\ Score_i = \frac{2 * Precision_i * Recall_i}{Precision_i + Recall_i}. \tag{20}$$

While IoU and MIoU are standard metrics for the assessment of semantic segmentation in computer vision research. The IoU is calculated for each class as follows, and its average over all photon classes defines the MIoU

$$IoU_i = \frac{TP_i}{TP_i + FP_i + FN_i}. \tag{21}$$

Bathymetric accuracy metrics quantify deviations between SDB predictions and in situ data. The RMSE, mean absolute error (MAE), median absolute error (MedAE), and goodness-of-fit for the 1:1 line (R^2) and the regression line (R_R^2) were used. The RMSE, particularly sensitive to outliers, is the most widely adopted metric for bathymetric studies, that is,

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (y_i - x_i)^2}{N}} \tag{22}$$

where x_i and y_i are the predicted and in situ bathymetry, respectively, for sample i, and N is the number of samples. In contrast, MAE and MedAE are more resilient against outliers. They are calculated as

$$MAE = \frac{\sum_{i=1}^{N} |y_i - x_i|}{N} \tag{23}$$

$$MedAE = median\left(|y_i - x_i|\right), i = 1, 2, 3, \ldots, N. \tag{24}$$

Finally, the goodness-of-fit metric is used to evaluate the statistical agreement between predictions and in situ water depths. R^2 is expressed relatively to the 1:1 line

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - x_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}, where\ \bar{y} = \frac{\sum_{i=1}^{N} y_i}{N}. \tag{25}$$

However, under specific environmental conditions, predictions may be correlated with in situ measurements while deviating from a 1:1 correspondence. In such cases, R_R^2 is used instead of R^2. It is expressed as

$$R_R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - f_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \tag{26}$$

where f_i is the fitting value for sample i based on the linear regression between x and y.

TABLE III
COMPARATIVE MODEL ACCURACY EVALUATION FOR PHOTON RECOGNITION WITH THE TESTING DATASET (↑ INDICATES THAT THE HIGHER THE VALUE, THE BETTER THE PERFORMANCE, AND VICE VERSA)

Model	MIoU↑	OA↑	F1 Score↑		
			Noise	Seafloor	Surface
DBSCAN	0.7810	0.9278	0.9276	0.7213	0.9549
AVEBM	0.3578	0.5627	0.4771	0.4481	0.6408
PointNet++	0.8297	0.9477	0.9260	0.8075	0.9744
DECOMPOSE	**0.9559**	**0.9867**	**0.9753**	**0.9623**	**0.9942**

B. Accuracy Evaluation of Photon Recognition With the Testing Dataset

Accurate photon event recognition, specifically the semantic segmentation of the ATL03 photon point cloud, is essential to the ICESat-2 SDB mission. Here, DBSCAN and AVEBM are training-free methods, whereas PointNet++ and DECOMPOSE require supervised learning with annotated training data prior to inference. Photon recognition results of DECOMPOSE and the baseline models are summarized in Table III.

DBSCAN attained an MIoU of 0.781 with an OA of 0.9278 on the testing dataset, yielding satisfactory though not optimal results. However, its $F1$-score for seafloor photons was only 0.7213, likely attributable to the markedly misclassifications caused by lower photon density at greater depths. Notably, as mentioned above, DBSCAN is not suited for practical scenarios because it does not output meaningful semantic labels. AVEBM was specifically designed for robustness against photon density variations caused by water depth. However, on the testing dataset, its performance was poor, with an MIoU of 0.3578 and a maximum $F1$-score of 0.6408. This is likely because the fixed AVEBM parametric configuration is adapted to short, curated segments with hierarchical photon distributions, but not to spatially heterogeneous photon densities observed along long ICESat-2 ground tracks. In contrast, PointNet++ substantially outperformed both DBSCAN and AVEBM across all metrics, achieving an MIoU of 0.8297 and a seafloor $F1$-score of 0.8075, which are significantly higher than the corresponding values obtained by DBSCAN (0.7213) and AVEBM (0.4481). These results emphasize the better performance of PointNet++ for seafloor photon recognition, demonstrating the suitability of deep learning models for ICESat-2 photon recognition. Finally, DECOMPOSE achieved the best performance, with an MIoU of 0.9559 (15.2% higher than PointNet++) and $F1$-scores consistently exceeding 0.96 (0.9623 for seafloor photons). This accuracy improvement demonstrates the excellent performance of DECOMPOSE.

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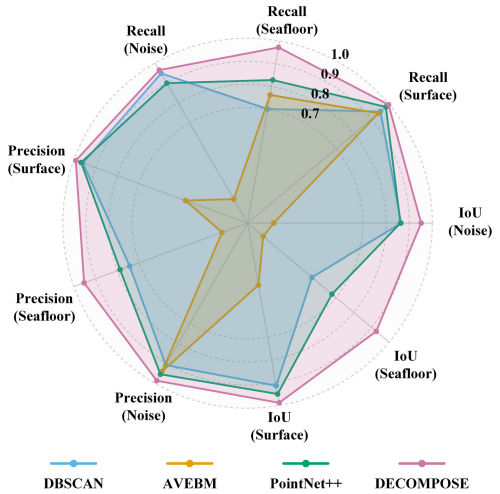


Fig. 8. Accuracy metrics for photon recognition: IoU, Precision, and Recall. The results for DECOMPOSE and the baselines are indicated separately for three semantic classes: seafloor, surface, and noise.

Fig. 8 compares class-specific performance in terms of IoU, precision, and recall, all of which are within the range [0,1], with higher values indicating better performance. AVEBM consistently underperformed for all semantic classes (IoU <0.5) when applied to the long-range ICESat-2 tracks, confirming its unsuitability for such complex environments. Although the overall performances of DBSCAN and PointNet++ were comparable, the seafloor class accuracy of PointNet++ was obviously higher, reflecting its stronger capacity for seafloor detection. Finally, DECOMPOSE yielded consistently better performances, with particularly notable gains over the baseline models for seafloor photon recognition: 0.25 for the IoU (reaching 0.9274) and 0.1 or more for precision and recall (both higher than 0.9). These results highlight the superiority of DECOMPOSE in ICESat-2 photon recognition, especially for seafloor photons.

To complement the analysis, DECOMPOSE performance was also qualitatively evaluated via visual interpretation. Fig. 9 presents the GT and photon recognition results for the four models over the Anegada site for a representative ICESat-2 transect. The site is characterized by homogeneous underwater terrain over tens of kilometers and water depths reaching below 25 m, providing low noise and favorable bathymetric visibility. It can be seen that DBSCAN achieved acceptable results [see Fig. 9(b)], but it misclassified seafloor photons as sea surface in shallow waters and as noise in deep waters, resulting in an MIoU of only 0.6337. The AVEBM performance was poor [see Fig. 9(c)], with near-complete classification failure across most of the transect, except in isolated shallow waters where it achieved limited accuracy. Although PointNet++ accuracy was markedly higher, with MIoU of 0.8297, a nonnegligible proportion of seafloor photons were misclassified [see Fig. 9(d)]. Finally, DECOMPOSE [see Fig. 9(e)] achieved high accuracy (MIoU of 0.9073), showing excellent agreement with the GT and minimal deviations, thus confirming its high usefulness and robustness against depth-induced photon density variability. Furthermore, to substantiate DECOMPOSE's superior performance, Appendix A presents a representative analysis of the Bora–Bora site, a lagoonal area markedly different from the relatively flat and continuous seafloor of Anegada. The obtained results are consistent with those observed in Fig. 9, indicating that DECOMPOSE maintains greater robustness across different environments. Overall, across all sites in the testing dataset, DECOMPOSE achieved substantial accuracy improvements compared to the baseline models, validating its efficacy and generalization capability.

C. Ablation Experiment

The key innovation of DECOMPOSE lies in its C-MIS-based cross-modality fusion paradigm. To quantitatively evaluate its importance and contribution to the recognition performance, we conducted an ablation experiment. An important consideration is that C-MIS units are functionally dependent on simultaneous input from both modality branches; thus, removing the imagery modality branch reduces DECOMPOSE to a transformer model composed exclusively of three PCTrans blocks, which served as the baseline in the ablation experiment. Alternatively, removing the C-MIS units fixes the channel dimension and can lead to a loss of feature richness. In conventional model design, the number of channels typically increases with network depth to facilitate abstract feature extraction, which was not the case for the PCTrans model used in the ablation experiment. To address this, "PCTrans with Rising Channel" was also implemented, a variation aimed at enhancing abstract feature extraction by increasing the number of channel dimensions, facilitating a comprehensive comparison.

For each of the models used in the ablation experiment, the average MIoU over four runs was calculated (see Fig. 10). First, DECOMPOSE achieved substantially higher accuracy, with a mean MIoU of 0.9540, than PCTrans (0.9201) and PCTrans with Rising Channel (0.8996). Second, it also demonstrated excellent consistency and minimal variability across multiple runs, as reflected by the compact error bars indicating a low standard deviation, whereas the pure transformer models exhibited considerable fluctuations in performance. Notably, increasing the number of feature channels degraded not only accuracy, but also its robustness, potentially resulting in recognition performance inconsistency between different runs. Although lower than DECOMPOSE, the accuracy of both transformer-based models remained substantially higher than that of the convolutional-kernel based PointNet++, highlighting the effectiveness of self-attention mechanisms for point cloud processing. Overall, the ablation experiment demonstrates the effectiveness of the multiscale photon feature fusion provided by C-MIS, further confirming the cross-modality fusion as a core component of DECOMPOSE.

D. Bathymetric Accuracy Evaluation for DECOMPOSE With the Real-World Dataset

The real-world dataset here designed covers a broader spatial scale than the training and testing datasets and conditions

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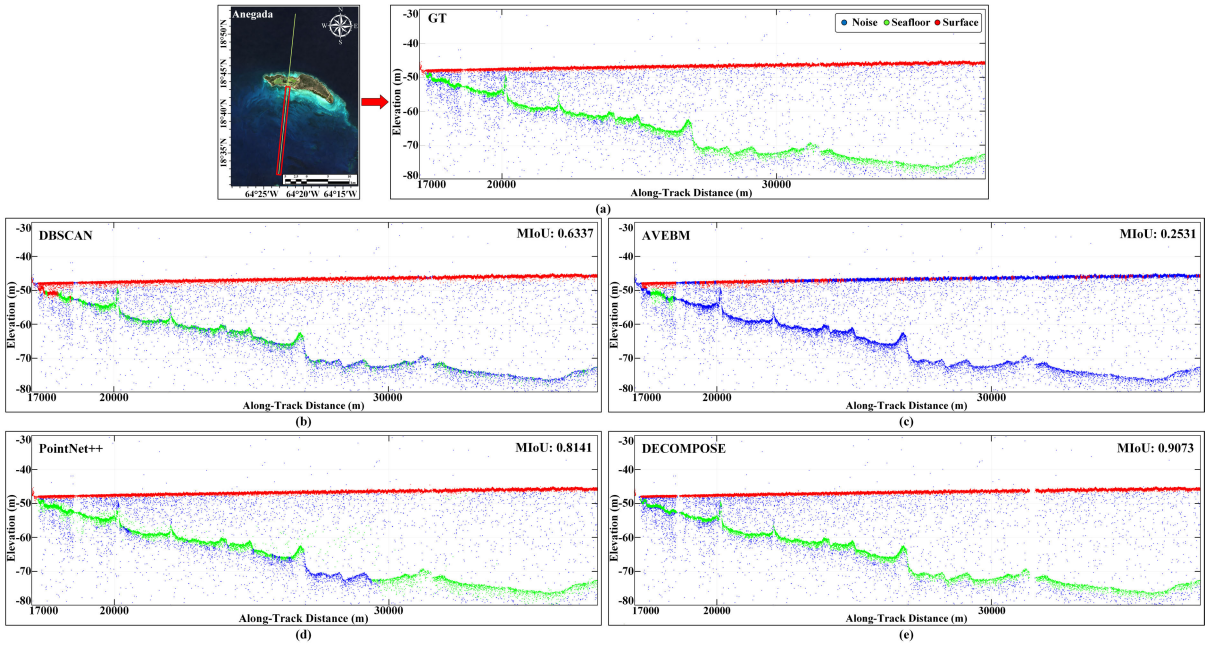


Fig. 9. Example of photon recognition over the Anegada site of the testing dataset for one ICESat-2 ground track transect. (a) Image of Anegada Island and annotated ATL03 photon point cloud (GT). The photon recognition results are illustrated for the baseline models (b) DBSCAN, (c) AVEBM, and (d) PointNet++, and for (e) proposed DECOMPOSE model.

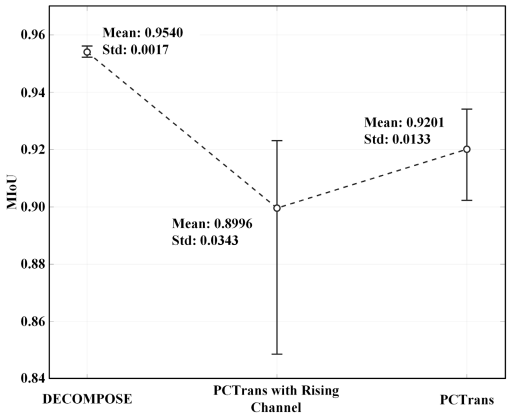


Fig. 10. Ablation experiment results (mean MIoU, open circles) for DECOMPOSE, PCTrans (backbone of the point cloud modality branch), and a modified PCTrans version with more channels to enhance abstract feature extraction (PCTrans with rising channel). Error bars extend between the minimum and maximum MIoU values over four runs.

(water IOPs and seafloor sediment composition), whose variations induce a pronounced data distribution shift. This dataset constitutes an exemplary OOD benchmark, suited to fulfill the fundamental objective of SDB: mapping underwater topography in regions with complex environmental backgrounds and inaccessible to in situ surveys. In this section, this dataset was used to assess the generalizability of DECOMPOSE. In the real-world dataset, the GT consists of water depths derived from in situ data, rather than manually annotated semantic labels. We evaluated the DECOMPOSE bathymetric accuracy for each site and for the entire dataset, with results summarized in Table IV.

The lowest accuracy was obtained for the Manu'a Islands, with an RMSE of 1.0712 m and an R^2 value of 0.7664,

TABLE IV
DECOMPOSE BATHYMETRIC ACCURACY EVALUATION WITH THE REAL-WORLD DATASET

Site	RMSE↓ (m)	MAE↓ (m)	MedAE↓ (m)	R^2↑	R^2_R↑
Niihau	0.7273	0.4839	0.3468	0.9714	0.9737
Manu'a Islands	1.0712	0.8809	0.8283	0.7664	0.8403
Florida Keys	0.4575	0.3464	0.2638	0.9679	0.9748
Puerto Rico	0.3321	0.2477	0.1942	0.9974	0.9974
Total	0.4938	0.2760	0.2575	0.9872	0.9881

indicating considerable deviation from reference in situ data. Conversely, the best performance was achieved for Puerto Rico, with a particularly low RMSE of 0.3321 m and near-optimal results for R^2 and R^2_R, both exceeding 0.99. For the Florida Keys and Niihau, the DECOMPOSE performance was comparable. Fig. 11 illustrates the deviations between model predictions and in situ GT data. Overall, there is clearly strong agreement between the DECOMPOSE predictions and in situ measurements, with R^2 values consistently above 0.96 for three sites, but not for the Manu'a Islands, where linear correlations between predictions and in situ data are relatively weaker (R^2 = 0.7664). Despite the environmental complexity and geographic diversity covered by the real-world dataset, the proposed DECOMPOSE achieved high accuracy and exhibited generalizability across domains, with RMSE and R^2 of 0.4938 m and 0.9872, respectively. To complement the quantitative performance assessment, a visual evaluation of seafloor terrain recognition was also performed. For each site of the real-world dataset, Fig. 12 exhibits a background satellite image with the selected ICESat-2 transect (left), the corresponding raw photon point cloud data (upper right), and the comparison (bottom right) between the terrain recognition results from

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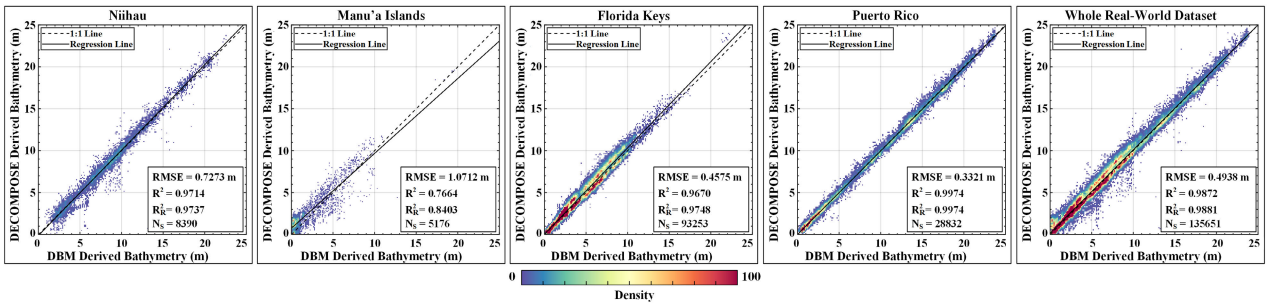


Fig. 11. Density scatter plots illustrating the relationship between DECOMPOSE bathymetry results and GT values derived from in situ DEMs, at each site and for the complete real-world dataset.

DECOMPOSE and from in situ data. All elevations were referenced to the WGS84 ellipsoid datum. The signal-to-noise ratio (SNR) metric was calculated for each transect to quantify the impact of noise on performance. Here, SNR is defined as the ratio of the number of valid oceanic photons N_S (confidence level >1 for ocean type) to the number of nonoceanic photons N_N

$$SNR = \frac{N_S}{N_N}. \tag{27}$$

A higher SNR value indicates fewer noisy photon signals and, therefore, higher data quality.

At the Niihau site [see Fig. 12(a)], there was strong consistency between predictions, i.e., photons classified as seafloor by DECOMPOSE (green dots), and in situ elevations (red dots). The calculated SNR (11.0222) indicated moderately low noise levels, hence satisfactory data quality. Over the Manu'a Islands [see Fig. 12(b)], DECOMPOSE accuracy was the lowest among real-world datasets. Despite a moderately high SNR value of 10.2873, obvious deviations occurred between the recognized seafloor and the reference in situ data. In particular, there are strong discrepancies in the marginal zones, where photon returns are sparse, which likely accounted for the large RMSE and low R^2 values (see Table IV). At the Florida Keys site [see Fig. 12(c)], the SNR was 15.4932, and both seafloor and surface photon clusters exhibited clear and robust structural continuity. This high-quality data allowed the DECOMPOSE to capture underwater topographic variations along the complete transect (approximately 10 km) with high accuracy and close agreement with the in situ data. In contrast, observations along the transect from Puerto Rico [see Fig. 12(d)] were affected by considerable scattering from the atmosphere and the water column, resulting in a low SNR value of 4.134. Nevertheless, the DECOMPOSE seafloor photon recognition performance remained robust, in close agreement with in situ data despite substantial noise in the raw point cloud. Notably, the model successfully mapped the seafloor, even under fragmented photon point cloud conditions shown in Fig. 12(b)–(d).

By combining quantitative and qualitative analyses, the superior performance of DECOMPOSE and its feasibility for practical deployment are validated on the real-world dataset. During cross-regional extrapolation, DECOMPOSE effectively handled OOD challenges caused by heterogeneity of IOP and benthic substrates. This confirms its strong generalizability and operational feasibility across diverse marine environments, fulfilling the essential practical requirements for the SDB approach. However, it remained constrained by the sparse distribution of ICESat-2 ground tracks and the resulting limited bathymetric coverage over large geographic regions.

E. Full-Coverage Bathymetric Accuracy Evaluation of DAPF-SDB Within the Real-World Dataset

To alleviate the constraints induced by ICESat-2 transect sparsity, the DAPF-SDB paradigm integrates ICESat-2 ATL03 point cloud data with Sentinel-2 L2A SR imagery. In particular, it uses bathymetric control points derived by DECOMPOSE from ICESat-2 to fine-tune the SR imagery based MuSRFM model for full-coverage underwater terrain mapping at depths of 0–25 m without in situ data. To evaluate, in a controlled manner, the effectiveness of the DAPF-SDB, which relies on a fine-tuned MuSRFM for full-coverage bathymetry, LBR, RFR, and CatBoost are employed as baseline models. These models use identical DECOMPOSE-derived bathymetric control points for regression fitting and subsequent full-coverage mapping. We further compare DAPF-SDB with the extrapolation results of the pretrained MuSRFM without fine-tuning to assess the contribution of DECOMPOSE to this paradigm.

Table V summarizes the performance of DAPF-SDB and the baselines across four experimental regions. Overall, all used models exhibited consistent performance across the four sites, achieving the highest performance in the Florida Keys, with accuracy progressively degrading at Puerto Rico, Niihau and Manu'a Islands. LBR performed worst across all sites, being highly sensitive to sparse ICESat-2 tracks and failing in Manu'a Islands, where its RMSE reached 19.5560 m, and R^2 became strongly negative (with a value of −5.5931). The machine learning based baselines, namely RFR and CatBoost, performed better and exhibited similar behaviors, achieving high accuracy in the Florida Keys but experiencing substantial degradation under OOD conditions, particularly in Manu'a Islands. In particular, CatBoost and RFR achieved RMSE of 0.8569 and 0.9484 m in the Florida Keys, respectively, but their RMSE increased markedly to 8.0531 and 8.2669 m in Manu'a Islands, indicating a dramatic variation in performance. Nevertheless, MuSRFM, despite requiring no fine-tuning, maintained strong and stable performance under challenging extrapolation scenarios, without dramatic collapse of accuracy or robustness. Analysis of the results showed that, except in the Florida Keys, where its RMSE was slightly lower than that of CatBoost, MuSRFM consistently outperformed LBR, RFR, and CatBoost across the other sites. In particular, in the Manu'a Islands, its

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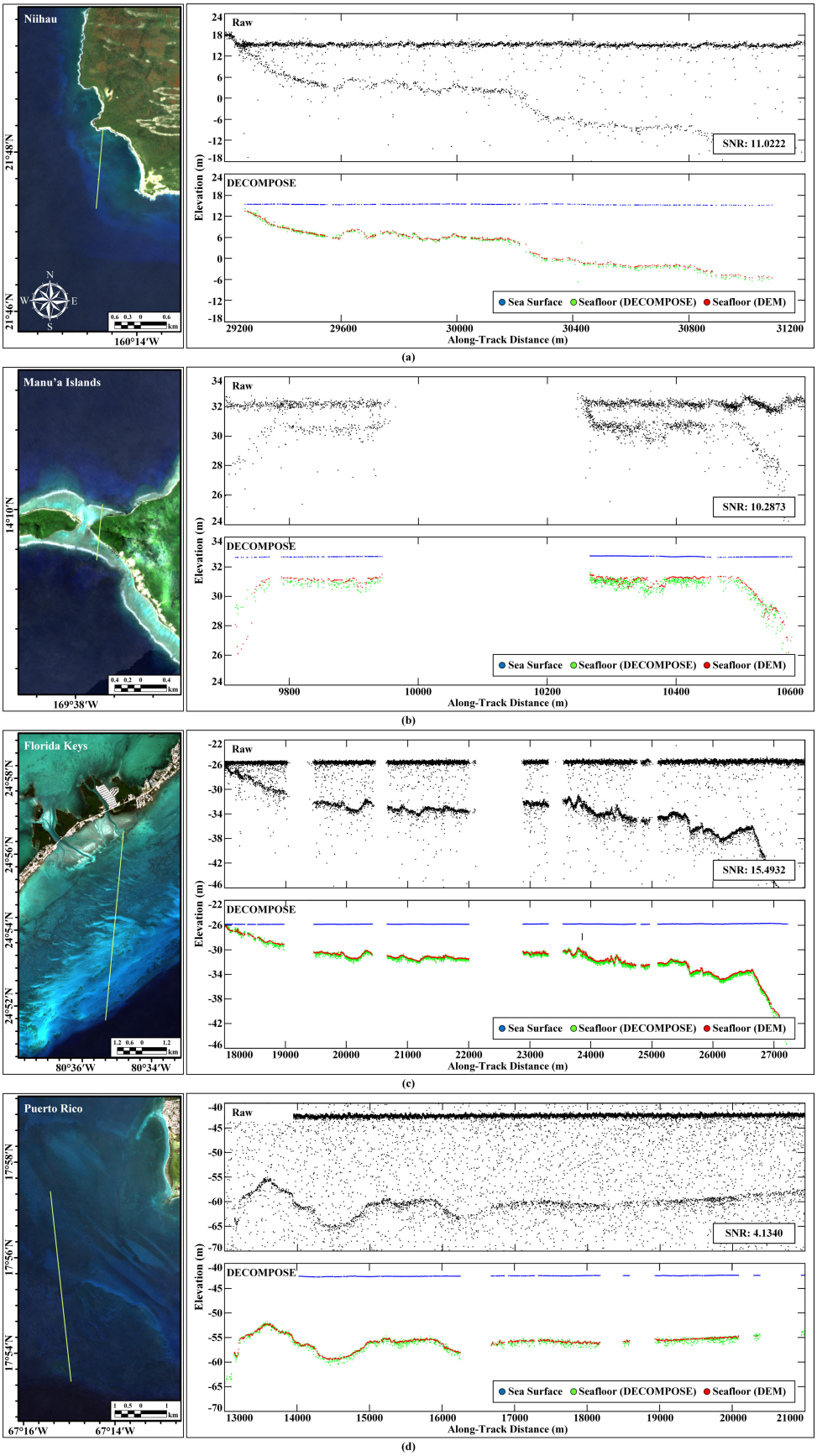


Fig. 12. Comparison of DECOMPOSE seafloor photon recognition with in situ elevation measurements for single ICESat-2 transects at real-world dataset sites: (a) Niihau, (b) Manu'a Islands, (c) Florida Keys, and (d) Puerto Rico.

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TABLE V

BATHYMETRIC ACCURACY EVALUATION FOR MUSRFM AND THE DAPF-SDB MODEL WITH THE REAL-WORLD DATASET († INDICATES THAT THE MODEL WAS NOT FINE-TUNED USING THE BATHYMETRY DERIVED FROM DECOMPOSE)

Site	Model	RMSE↓ (m)	MAE↓ (m)	MedAE↓ (m)	R^2↑	R^2_R↑
Niihau	**LBR**	4.0794	2.9786	2.0146	0.5595	0.6755
	RFR	3.8547	2.6105	1.7268	0.6067	0.6452
	CatBoost	3.9680	2.6861	1.6686	0.5832	0.6277
	MuSRFM†	1.9573	1.5671	1.3423	0.8935	0.8946
	DAPF-SDB	**1.5076**	**1.1255**	**0.8502**	**0.9368**	**0.9390**
Manu'a Islands	**LBR**	19.5560	4.1057	1.8458	-5.5931	0.0354
	RFR	8.2669	5.5973	2.4605	-0.1782	0.2152
	CatBoost	8.0531	5.4063	2.2061	-0.1180	0.2513
	MuSRFM†	3.1749	2.7506	2.6671	0.7991	**0.9297**
	DAPF-SDB	**2.6378**	**1.9072**	**1.3747**	**0.8613**	0.8920
Florida Keys	**LBR**	1.1861	0.8473	0.6615	0.8406	0.8423
	RFR	0.9484	0.6339	0.4034	0.8981	0.9106
	CatBoost	0.8569	0.5708	0.3758	0.9168	0.9288
	MuSRFM†	0.9015	0.7741	0.7697	0.9091	0.9412
	DAPF-SDB	**0.6713**	**0.4903**	**0.3446**	**0.9496**	**0.9533**
Puerto Rico	**LBR**	2.8393	2.2786	2.0630	0.7473	0.7511
	RFR	2.3803	1.7633	1.3223	0.8224	0.8419
	CatBoost	2.3844	1.7564	1.2933	0.8218	0.8425
	MuSRFM†	1.8045	1.3602	1.0640	0.8992	0.9132
	DAPF-SDB	**1.5032**	**1.0486**	**0.7043**	**0.9301**	**0.9351**

RMSE was 3.1749 m, more than 60% lower than that obtained by CatBoost, which was calibrated using DECOMPOSE-derived bathymetry. This result demonstrates that MuSRFM exhibits superior generalization capability when applied to complex unseen data. Finally, as anticipated, DAPF-SDB, integrating DECOMPOSE and MuSRFM, delivered superior performance, significantly surpassing all baselines across the entire real-world dataset. Quantitatively, DAPF-SDB achieved substantial performance gains, with RMSE, MAE, and MedAE reduced by more than 16%, more than 14%, and more than 8%, respectively, and R^2 increased by more than 3%. The slightly lower R^2_R obtained for DAPF-SDB in Manu'a Islands is anomalous and, given the substantial gains in the other metrics, does not reflect its general performance. Overall, the results demonstrate that DAPF-SDB's accuracy arises from the complementary roles of DECOMPOSE and MuSRFM. Notably, bathymetric control points derived from the DECOMPOSE, adapting the MuSRFM to the data distribution of the target site and obviating the need for in situ measurements.

As noted earlier, despite differences in absolute accuracy, all models exhibit a consistent highest-to-lowest accuracy ranking: Florida Keys, Puerto Rico, Niihau, and Manu'a Islands. The performance discrepancy between these sites is likely due to geomorphological differences. The Manu'a Islands site features steep volcanic underwater slopes, which challenge bathymetric retrievals, whereas the Florida Keys site is characterized by gentle and shallow terrain, favorable for SDB. The following paragraphs present a site-specific evaluation of DAPF-SDB performance to enable a closer examination of each site.

Fig. 13 illustrates the results at Niihau, the westernmost and smallest of the inhabited Hawaiian Islands. The island's sparse population and geographic isolation limit the influence of terrigenous suspended matter on its optical environment. It can be seen that there was good general consistency between the DAPF-SDB's predictions and DBM [see Fig. 13(a)], corroborated by dense clustering along the 1:1 line in the density scatter plot [see Fig. 13(d)]. However, closer examination of small regions with finer-scale topographic details indicated limitations in the DAPF-SDB resolution ["blurred" aspect of the DAPF-SDB bathymetric map, Fig. 13(b) and (c)]. At Kawa Point [see Fig. 13(b)] and Lehua Island [see Fig. 13(c)], large-scale underwater topographic structures are accurately captured, demonstrating the model's ability to represent key geomorphological features. However, fine-scale details, particularly in deeper-water regions, are less accurate due to steep slopes and limited optical penetration.

Fig. 14 illustrates the results for the Manu'a Islands, the site where the proposed DAPF-SDB model achieved the lowest accuracy among the real-world datasets. The steep underwater terrain around this volcanic archipelago considerably restricted the detectable bathymetry to a narrow near-shore region [see Fig. 14(a)]. The enlarged views [see Fig. 14(b) and (c)] indicate clear systematic deviations between the DAPF-SDB predictions and in situ data, with an overestimation of the bathymetry by the model in shallow-water areas and an underestimation in deep-water areas. This depth-dependent bias explains the strong deviation of the regression line from the 1:1 line shown in Fig. 14(d) and can be attributed to the low bathymetric capability of MuSRFM under steep underwater topography conditions. The sparse density distribution in Fig. 14(d) reflects the suboptimal performance of DAPF-SDB under these environmental conditions, consistent with the high RMSE and low R^2 reported in Table V.

The Florida Keys, located south of the Florida Peninsula (see Fig. 15), are surrounded by extensive littoral shallow waters with limited deep-water regions. Here, the DAPF-SDB bathymetry achieved excellent accuracy and consistency with the in situ DBM [see Fig. 15(a)], confirmed by the

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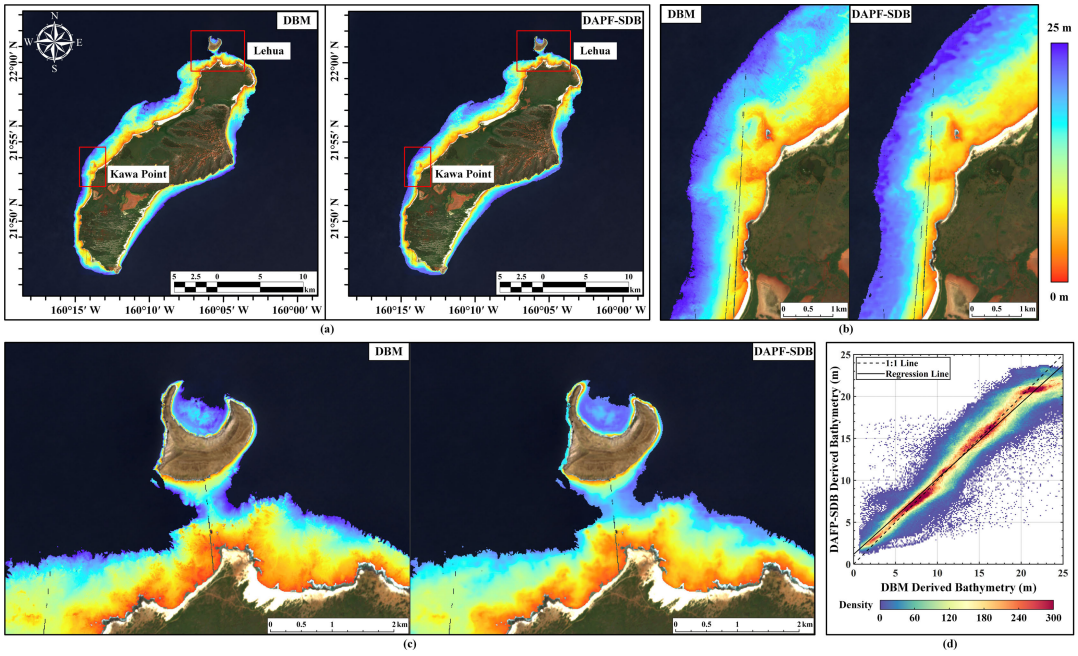


Fig. 13. Bathymetric map comparison (a) between an in situ DBM and DAPF-SDB for the real-world dataset at the Niihau site. Finer-scale comparison is illustrated around (b) Kawa Point and (c) Lehua Island. (d) Density scatter plot of the relationship between the DBM and the DAPF-SDB predictions.

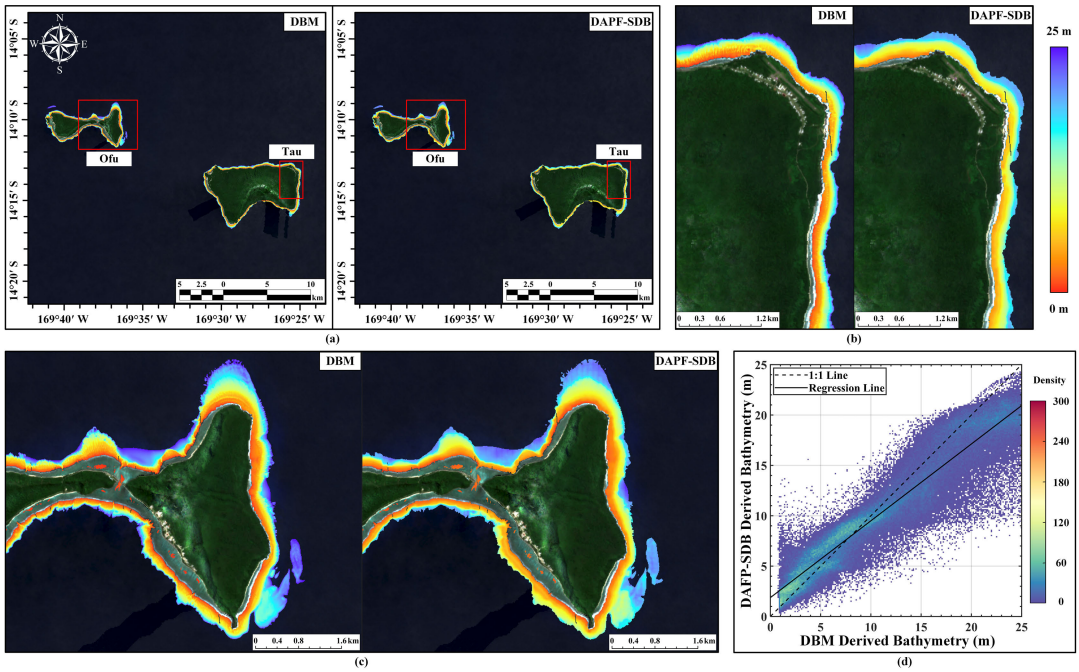


Fig. 14. Bathymetric map comparison (a) between an in situ DBM and DAPF-SDB for the real-world dataset at the Manu'a Islands site. Finer-scale comparison is illustrated around (b) Tau and (c) Ofu. (d) Density scatter plot of the relationship between the DBM and the DAPF-SDB predictions.

quantitative metrics reported in Table V: extremely low RMSE of 0.6713 m, and R^2 and R_R^2 reaching 0.9496 and 0.9533, respectively. Around Watsons Reef [see Fig. 15(b)], DAPF-SDB produced accurate bathymetric predictions down to 20 m, albeit with partial loss of fine-scale morphological details in deep waters far from the coast. Around Lower Matecumbe Key [see Fig. 15(c)], the DAPF-SDB produced a shallow-water bathymetric map of excellent quality, accurately resolving complex coastal hydrodynamic landforms and estuarine features. The dense point clustering along the 1:1 line in Fig. 15(d) further supports these findings.

In Puerto Rico (see Fig. 16), DAPF-SDB produced generally accurate bathymetry in strong agreement with in situ DBM data [see Fig. 16(a)]. In this area, depth gradually increases toward the open sea, forming a typical near-shore-to-offshore subaquatic topographic gradient. The enlarged views around Bajo Casabe and Arrecife Peregrina (see Fig. 16(b) and (c), respectively) further highlight systematic DAPF-SDB biases,

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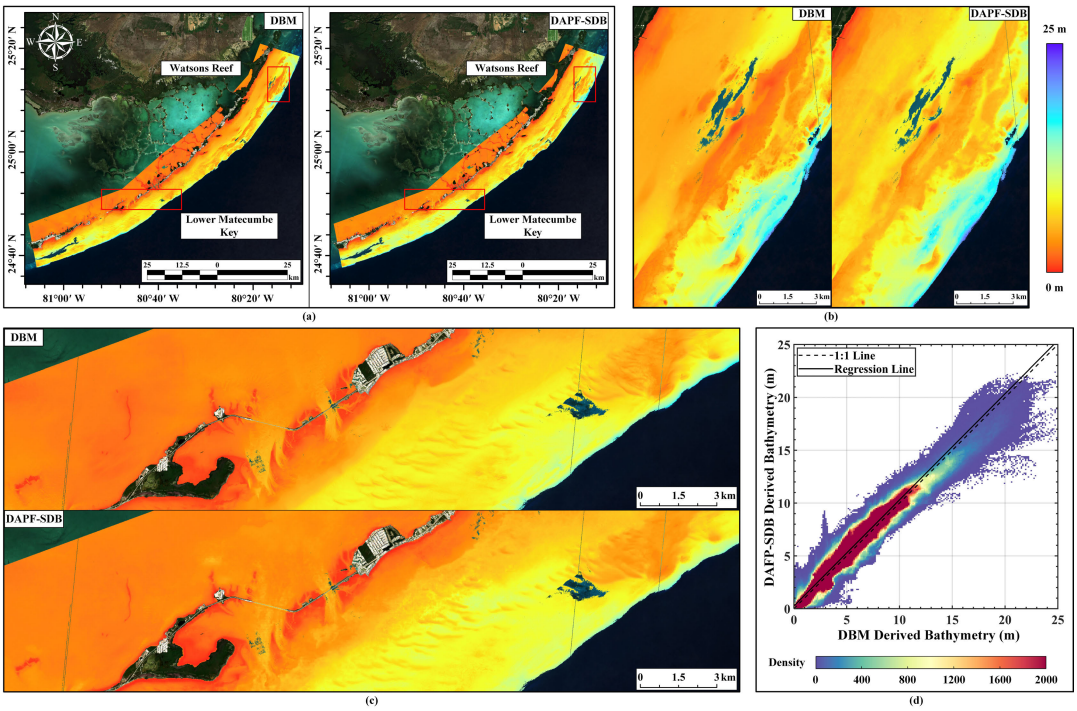


Fig. 15. Bathymetric map comparison (a) between an in situ DBM and DAPF-SDB for the real-world dataset at the Florida Keys site. Finer-scale comparison is illustrated around (b) Watson's Reef and (c) Lower Matecumbe Key. (d) Density scatter plot of the relationship between the DBM and the DAPF-SDB predictions.

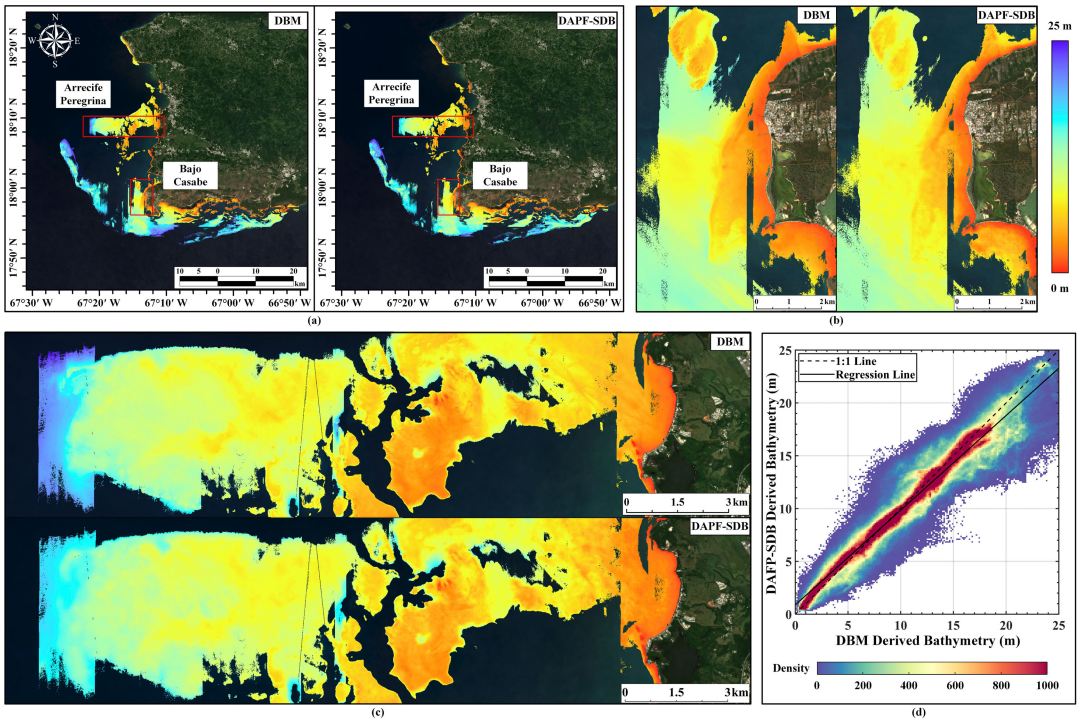


Fig. 16. Bathymetric map comparison (a) between an in situ DBM and DAPF-SDB for the real-world dataset at the Puerto Rico site. Finer-scale comparison is illustrated around (b) Bajo Casabe reefs and (c) Arrecife Peregrina. (d) Density scatter plot of the relationship between the DBM and the DAPF-SDB predictions.

similar to those observed at the Manu'a Islands site. These biases include shallow-water overestimation and deep-water underestimation, as corroborated by the regression line deviation from the 1:1 line in Fig. 16(d). A comparison of the scatter density plots reveals that this site exhibits greater dispersion and a larger spread of deviations, suggesting that spaceborne bathymetric mapping in this site is more challenging.

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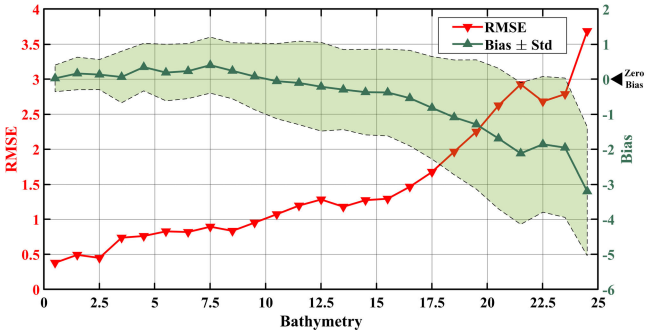


Fig. 17. DAPF-SDB prediction error relative to the in situ data is evaluated over the entire real-world dataset, covering the full range of water depths. The left ordinate corresponds to RMSE (red), while the right ordinate corresponds to bias (green). All values are in meters.

Fig. 17 depicts the error characteristics of DAPF-SDB across the full water-depth range on the entire real-world dataset. Both RMSE and bias exhibit an increasing trend with bathymetry. The bias and its associated standard deviation indicate that DAPF-SDB's performance is optimal at depths shallower than 15 m, whereas the pronounced degradation in deeper depths reflects substantially higher variability in the prediction. This tendency is likely attributable to the progressive attenuation of seafloor reflectance with increasing depth, which weakens its contribution to the observed spectral signal and consequently reduces the stability of the bathymetry-inversion mapping.

In summary, quantitative and qualitative assessments of the extrapolation capability of DAPF-SDB confirmed its accurate and robust full-coverage bathymetry without reliance on in situ data. Across four geographically diverse sites, DAPF-SDB yielded substantial improvements over baselines, verifying the effectiveness of its core components. While some limitations remain in reproducing fine morphological details, they do not undermine the general accuracy and robustness of DAPF-SDB.

V. Discussion

The experimental results shown in Section IV demonstrate DECOMPOSE's excellent performance in photon recognition and bathymetric prediction, and validate the improvement in full-coverage bathymetric mapping achieved by the synergistic DAPF-SDB model. Nevertheless, there remain limitations to the proposed methodologies, discussed in this section.

A. Influence of Anomalous Noise on DECOMPOSE Photon Recognition

ICESat-2 observations are frequently contaminated by suspended particulates and gas bubbles that backscatter laser pulses, producing noise signals that interfere with valid seafloor and sea surface reflected photons. The most common sources of noisy signals include suspended particulate matter, benthic gas seepage, or large aggregates of biological scatterers such as fish or shrimp; however, real situations can be more complex (e.g., include a combination of interference sources). Under specific environmental conditions, this anomalous noise can induce a marked increase in the photon

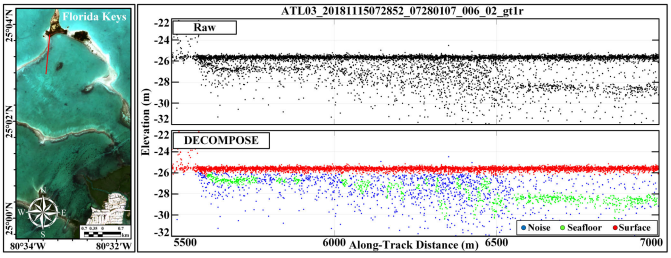


Fig. 18. Photon recognition for an ICESat-2 ground track transect with anomalous noise at the Florida Keys site (left), raw photon point cloud (top right), and DECOMPOSE prediction (bottom right). A substantial number of noise photons in the middle transect section were misclassified as seafloor.

density, substantially altering the apparent photon distribution. Fig. 18 illustrates this phenomenon along an ICESat-2 transect in the Florida Keys, where a high density of anomalous water-column noise signals is present. In this case, most photons in the central half of the transect were misclassified as seafloor photons. Such noises can induce vulnerabilities and extrapolation failures in SDB models that depend on density characteristics for photon recognition, including DECOMPOSE. Although these anomalous events were infrequent in our experiments, similar or even more severe noise contamination can be expected when the model is used at larger scales. Therefore, improving the resilience of SDB models against extreme noise is essential to ensure their reliable performance for practical applications. Two common solutions to solve this problem warrant further investigation: increasing the training dataset volume to better represent noise diversity or developing robust discriminative algorithms with reduced sensitivity to background noise interference.

B. Bathymetric Conversion Error

Although the proposed method does not require in situ data for training or inference, it relies on the DTU23 tidal model for bathymetric output and is therefore not entirely independent of external geophysical information. In addition, errors induced by refraction correction may also propagate and accumulate during this bathymetric conversion. The impact of this conversion error was evaluated using seafloor elevations referenced to the WGS84 ellipsoid datum by quantifying discrepancies between DECOMPOSE predictions and in situ measurements. This approach bypasses the bathymetric conversion, allowing direct quantification of error induced by this procedure.

Table VI quantifies the accuracy of DECOMPOSE-derived seafloor elevations, indicating a statistically significant gain in bathymetric accuracy reported in Table IV. Most notably, the total RMSE was 0.3497 m, approximately 29% smaller than the bathymetry-based RMSE (0.4938 m; Table IV). Site-specific RMSEs were reduced by 7%–39%, with other metrics also indicating substantial improvement, which demonstrates that the results of DECOMPOSE without bathymetric conversion are more accurate. Furthermore, compared to the bathymetric results shown in Fig. 11, the DECOMPOSE-derived seafloor elevations in Fig. 19 align more closely with the in situ GT, exhibiting lower dispersion and markedly fewer outliers. In particular, the goodness-of-fit based on elevation

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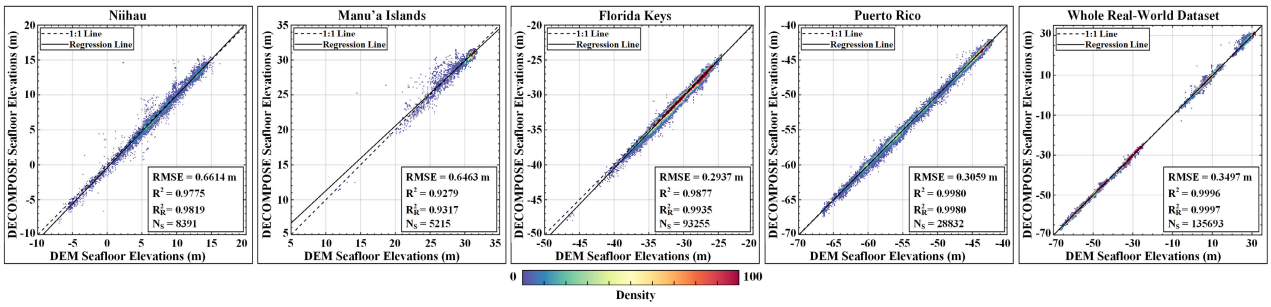


Fig. 19. Density scatter plots illustrating the relationship between seafloor elevations from DECOMPOSE and from in situ DEM, at each site and for the complete real-world dataset.

TABLE VI

ACCURACY EVALUATION OF DECOMPOSE SEAFLOOR ELEVATIONS WITH THE REAL-WORLD DATASET

Site	RMSE↓ (m)	MAE↓ (m)	MedAE↓ (m)	R^2↑	R^2_R↑
Niihau	0.6614	0.4391	0.3223	0.9775	0.9819
Manu'a Islands	0.6463	0.4156	0.3052	0.9279	0.9317
Florida Keys	0.2937	0.2129	0.1589	0.9877	0.9935
Puerto Rico	0.3059	0.2108	0.1439	0.9980	0.9980
Total	0.3497	0.2342	0.1655	0.9996	0.9997

achieved at the Manu'a Islands was notably higher than that obtained after bathymetric conversion, indicating that this conversion process substantially affects the final accuracy.

In-depth analyses identified systematic biases arising from two critical conversions: photon-to-bathymetry and DEM-to-DBM. These biases were attributed to three main error sources.

1) *Dependence on External Model:* The proposed in situ-free method relies on the DTU23 model for bathymetric referencing, and residual errors in marginal or poorly constrained regions inevitably bound the accuracy.
2) *Sea Surface Recognition Inaccuracies:* Vertical biases stem from offsets in the datum defined by sea surface photon detection, thus degrading the accuracy of the derived bathymetry.
3) *Inaccuracies of DBM:* The DBM uses the DTU21 MSS as a datum, and errors may arise from the global model itself or MSS interpolation during DEM-to-DBM conversion, rather than from the proposed method.

Currently, we cannot fully characterize these biases or their contributions, which will be addressed in future studies.

C. Propagation of DECOMPOSE Bathymetric Errors Into DAPF-SDB

By construction, the proposed DAPF-SDB model applies bathymetric values from DECOMPOSE as control points to adjust the MuSRFM weights. Consequently, errors included in the DECOMPOSE output can propagate into DAPF-SDB and degrade the overall performance. To assess the bias introduced by DECOMPOSE bathymetric errors and evaluate its impact on the final predictions, the Manu'a Islands site, where DAPF-SDB accuracy was lowest, was selected as the experimental site. In particular, DECOMPOSE-derived bathymetric values

TABLE VII

BATHYMETRIC ACCURACY COMPARISON BETWEEN MUSRFM, DAPF-SDB, AND GT FINE-TUNED MUSRFM USING REAL-WORLD DATA FROM THE MANU'A ISLANDS SITE

Model	RMSE↓ (m)	MAE↓ (m)	MedAE↓ (m)	R^2↑	R^2_R↑
MuSRFM	3.1749	2.7506	2.6671	0.7991	**0.9297**
DAPF-SDB	2.6378	1.9072	1.3747	0.8613	0.8920
GT Fine-tuned MuSRFM	**2.2511**	**1.6457**	**1.2380**	**0.8990**	0.9217

were replaced with error-free in situ DBM used as control points to fine-tune MuSRFM weights, generating a "GT Fine-tuned MuSRFM." Replacing these bathymetric control points with in situ data eliminates errors intrinsic to DECOMPOSE, thereby isolating and quantifying the impact of prediction errors on the nominal DAPF-SDB performance.

Table VII lists the bathymetric accuracy evaluation results for MuSRFM alone, the nominal DAPF-SDB configuration, and the GT fine-tuned MuSRFM at the Manu'a Islands site. Overall, all used metrics except R^2_R improved considerably. In particular, the RMSE decreased from 2.6378 m for DAPF-SDB to 2.2511 m for the GT fine-tuned MuSRFM, confirming the impact of DECOMPOSE prediction errors on the bathymetric accuracy of DAPF-SDB. However, despite the improvement after fine-tuning, the RMSE remained above 2.2 m, suggesting additional underlying error sources. Conversely, the results for R^2_R were inconsistent: the R^2_R obtained for the GT Fine-tuned MuSRFM (0.9217) was slightly lower than the value of 0.9297 obtained for MuSRFM alone. This anomaly is likely caused by fine-tuning, which adjusted the model weights to the statistical characteristics of the target domain, slightly perturbing the optimal linear relationships established in the source domain. Appendix B provides a more intuitive illustration through a comparison of the corresponding density scatter plots. In summary, the degradation of DAPF-SDB relative to GT Fine-tuned MuSRFM confirms the influence of errors induced by recognition results of DECOMPOSE, while its improvement over MuSRFM extrapolation underscores the contribution of DECOMPOSE within this active–passive fusion paradigm.

VI. CONCLUSION

The purpose of SDB is to obtain bathymetric information in remote or hydrologically complex regions where in situ

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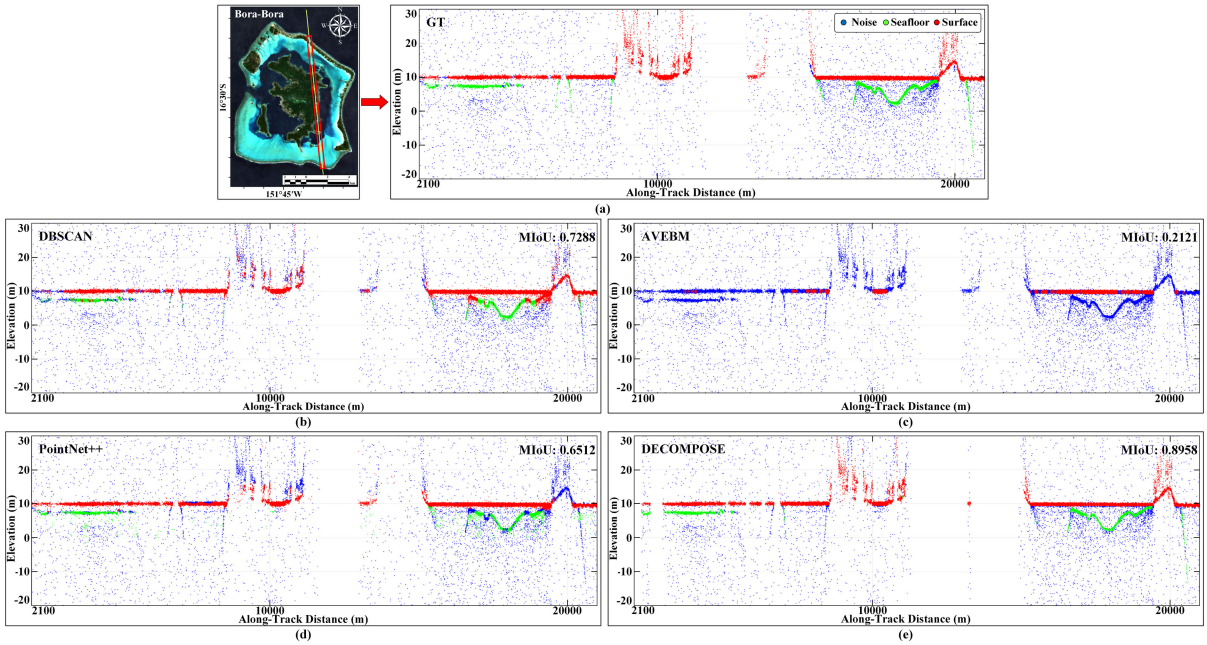


Fig. 20. Example of photon recognition over the Bora–Bora site of the testing dataset for one ICESat-2 ground track transect. (a) Image of Anegada Island and annotated ATL03 photon point cloud (GT). The photon recognition results are illustrated for the baseline models (b) DBSCAN, (c) AVEBM, and (d) PointNet++, and for (e) proposed DECOMPOSE model.

data cannot be acquired. In such environments, heterogeneous water IOPs and benthic composition create data distribution shifts that necessitate models with robust OOD extrapolation. To address this challenge, we proposed a novel cross-modality fusion that considerably enhanced both the ICESat-2 photon recognition accuracy and its generalization capability, while minimizing SDB dependence on in situ data through synergistic integration with Sentinel-2 multispectral imagery.

This study focused on developing accurate and generalizable SDB methods for transparent near-shore shallow waters. Our primary objective was to develop an accurate and robust SDB model from ICESat-2 active LiDAR observations. As the core contribution of this study, DECOMPOSE introduces a novel cross-modality fusion strategy that integrates photon density profiles and point clouds. This model achieved an MIoU of 0.9559, improving photon recognition accuracy by at least 15.2% over established baseline models. On an OOD real-world dataset, DECOMPOSE achieved an RMSE of 0.4938 m, demonstrating adequate cross-regional generalizability. The second major innovation was the integration of DECOMPOSE with MuSRFM, a Sentinel-2 multispectral imagery-based SDB model, into a synergistic model called DAPF-SDB, which provides full-coverage bathymetric capability without requiring in situ measurements. On the real-world dataset, DAPF-SDB outperformed all baseline models, achieving RMSE values of 0.6713–2.6378 m across depths of 0–25 m.

However, our limited training dataset could not entirely capture the complexity and variety of real-world underwater scenarios. This created persistent representativeness problems, such as misclassification and failures in complex environments. We intend to address these limitations in further studies, with a focus on increasing the training data diversity and improving the feature extraction capabilities and robustness.

APPENDIX A

This appendix evaluates the photon recognition performance of DECOMPOSE and baseline models at the Bora–Bora site, which features typical lagoonal topography, as shown in Fig. 20. DBSCAN outperformed PointNet++, likely because the latter misclassified many shallow-water seafloor photons as noise. AVEBM, by contrast, failed along this complex transect and was largely unable to recognize seafloor-reflected photons. In contrast, DECOMPOSE consistently outperformed baseline models, avoided large-scale misclassifications, and exhibited only minor errors near the coastline, confirming its robustness under complex topographic conditions.

APPENDIX B

This appendix evaluates Section V-C configurations using density scatter plots. As shown in Fig. 21, DAPF-SDB reduced dispersion around the 1:1 line and improved RMSE and R^2 relative to MuSRFM extrapolation. However, its lower R_R^2 value reflected residual offset and slope deviations, pointing to remaining scale bias. The GT-fine-tuned MuSRFM exhibited compact scatter and near coincidence of the regression and 1:1 line, with improvements across most metrics. Overall, DAPF-SDB substantially improved accuracy without in situ data, although small systematic biases attributable to DECOMPOSE persisted when compared with the GT fine-tuned case.

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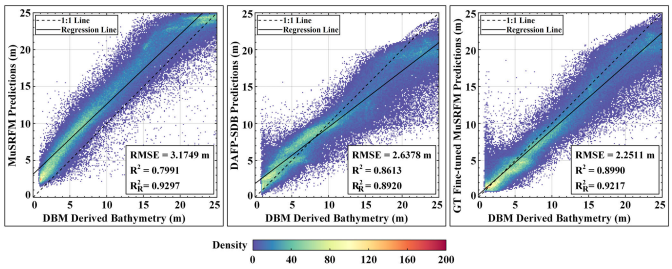


Fig. 21. Density scatter plots illustrating the relationship between model-derived bathymetry and in situ measurements for the MuSRFM (left), the proposed DAPF-SDB model (center), and the GT Fine-tuned MuSRFM (right) at the Manu'a Islands site of the real-world dataset.

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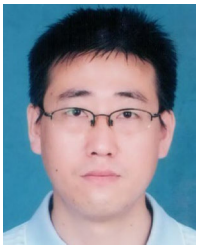
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